

PCB Design for Micromouse Robot

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ECE 499

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Report Summary

My project is to design and test a printed circuit board (PCB) for the micromouse robots made by the Union college robotics team (UCRC). A micromouse is a small (25x25cm maximum) autonomous robot that is designed to traverse a maze independently. Union college participates in a competition organized by IEEE every year where the winner is decided based on fastest completion time. Although previous iterations of micromice built by Union have performed well, they can be improved upon. Moving to a PCB based design will reduce some of the mechanical (weight, height/general size, which affects maneuverability), electrical and debugging pitfalls of last year's designs.

The mechanical requirements for this design are that it be 10cm x 7cm wide and 1.6mm thick. The material used to construct it will be FR-4 and only through-hole package components will populate it. The electrical requirements for the PCB are that it follows the schematic perfectly and that it will have a conductor spacing of 0.05mm and maximum trace width of 0.3mm. The planned budget was \$50 for the manufacturing of the PCB.

The design of the micromouse PCB went through three separate iterations. The third and final iteration PCB has six sensors at the front of the board, two motors in the middle, a microcontroller and motor controller stacked on top of each other, and a barrel jack to connect the motor controller battery. There are four spacers screwed into the board to hold up the platform that the microcontroller and motor controller batteries will be attached to. There are also ball casters installed at the front and back of the design for balancing.

Continuity tests were conducted on all PCBs. Only the second one failed because of issues with the left motors. The test codes were run to check the responsiveness of the IR sensors and motor encoders. Resulting encoder and IR sensors values were plotted against time. The plots showed that all the components operated properly and therefore proved the conductor spacing and trace width parameters used worked as predicted.

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1.0 Introduction and Project Definition

The micromouse competition is held by the IEEE every year. In this competition different colleges will build small autonomous robots that will solve a 16 x 16 block maze. These robots, referred to as micromice, will have to finish solving the maze within 10 minutes to win the competition. The competition will have judges that will pick a winner based on how fast the mouse completes its run. Besides the time taken, the judges will take into account the number of times the owner resets their stuck robot, the design of the robot and how well it performs in navigating the maze[1].

1.1 Problem definition

The Union College Robotics Crew is a participant in the IEEE micromouse competition and is planning to compete next spring. Currently there are five prototype micromice built by the UCRC which have several notable issues. One prototype, shown on *figure 1*, has a large Arduino Uno microcontroller mounted on top of it, making the micromouse top heavy. This caused the micromouse to tip forward whenever sudden stops were made. Adding to its weight is a three inch lipo battery mounted between the micromouse support beams. This heaviness occasionally stalls the micromouse motors, causing damage and depleting the battery. The convoluted wiring is also a significant problem. Even when color coded they cause great confusion provided any troubleshooting is required. In addition, these wires do not reliably stay stuck to their respective pins on the microcontroller because the micromouse is constantly put in motion. The wires also occasionally interfere with the IR sensors the micromouse uses to detect its surroundings, leading to further problems with the general performance of the micromouse. In addition it can be observed from *figure 1* that there are several improvisations in the prototype micromouse, such as the glued popsicle sticks that are meant to operate as bumpers. Reducing the height & weight, condensing the wiring and strengthening the connections can drastically improve the performance of the micromouse.

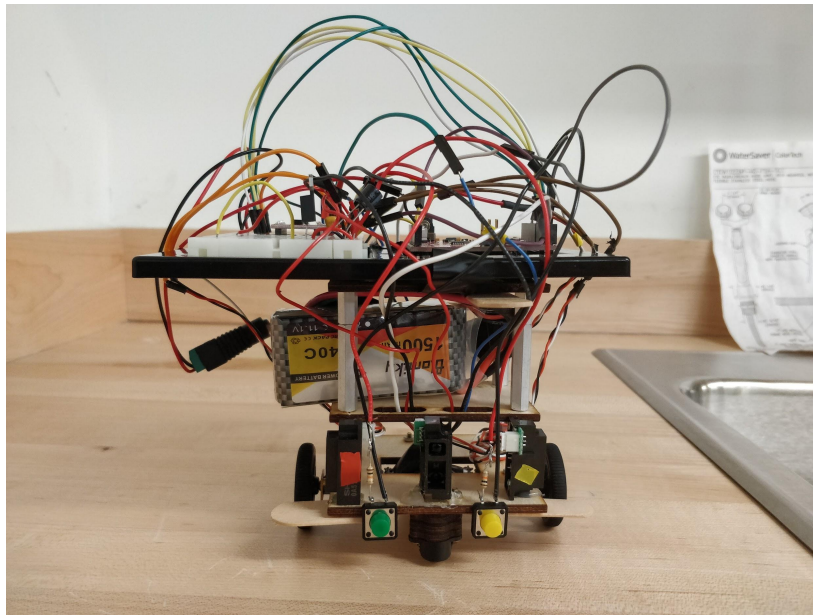


Figure 1. *Prototype UCRC Micromouse*

1.2 Project goal

The purpose of this project is to design a printed circuit board (PCB) to replace the breadboard circuit used in last year's designs. The PCB will serve as both the electrical circuitry and the mechanical chassis. This change will likely reduce or eliminate the many mechanical and electrical problems from previous micromouse designs. Mechanically, the PCB micromouse will be lighter and smaller than the previous robots. The reduced weight of the micromouse will decrease the load on the motors and thus avoid stalling (which was a major problem last year). The smaller size means the robot will have more space to move within a maze block, which will leave more room for error during sharp turns and will help the robot readjust itself more easily without crashing to any walls.

The lack of wire clutter also means the micromouse will be much easier to debug should the robotics team run into any issues. Every trace on the printed circuit board will be easily accessible to a multimeter probe and we can be sure that the design follows the schematic sent to the manufacturer. The last added benefit would be that the PCB micromouse will be long lasting. Most of the robots the UCRC has built in the past have broken down in recent years due

to loose and exposed wiring as well as use of temporary mechanical solutions (i.e tape & glue) to assemble micromouse components. As a result, these robots could not be used to train newer crew members. The PCB micromouse will not have this issue and can be used for years to come.

2.0 Background

Micromouse design began sometime in the 1970s when the IEEE spectrum magazine introduced the idea of a maze solving robot vehicle [2]. The magazine was apparently inspired by an earlier design of mechanical mice that were intended for racing. Only 15 micromice were allowed to compete out of 6000 entries. The winning micromouse, called ‘moonlight flash’, was a wall follower robot that did not use the complicated flood fill algorithms we use today [3] Since then the competition has gotten far more advanced. The fastest micromouse in the world at the moment can solve its maze in 3.9 seconds [4]

Here at Union there have been several attempts to create a PCB micromouse. A few years ago Edward Gebara created the PCB shown on *figure 2*. Most recently, Nikolaos Papaionnou, a member of the UCRC, attempted to design a PCB for one of Union’s micromice as well. However, his design never reached the manufacturing stage.

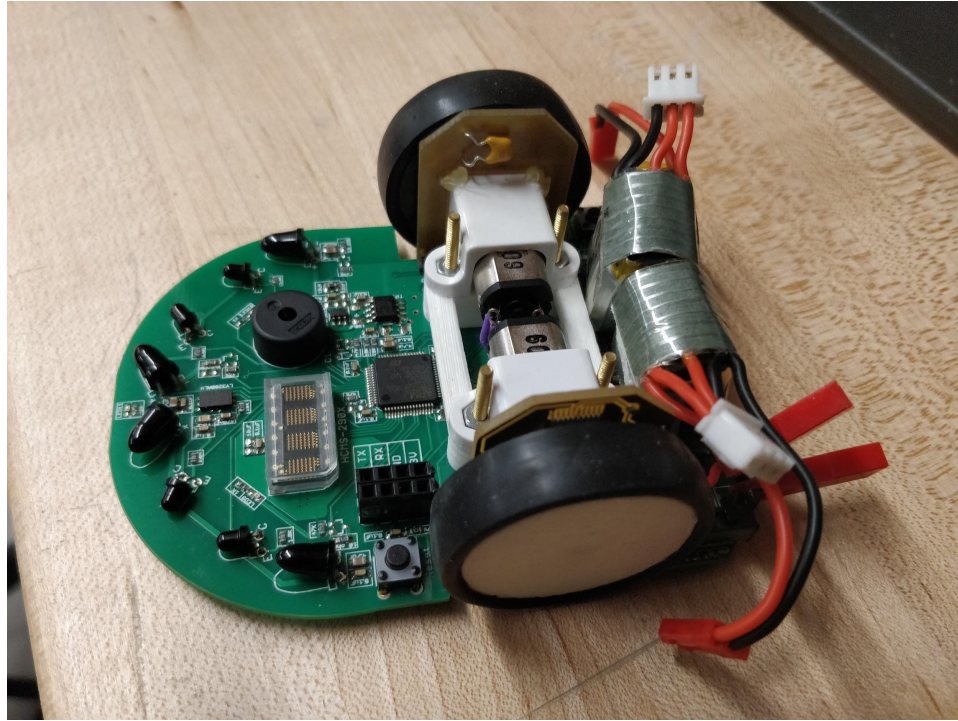


Figure 2. *Previous PCB Micromouse design at Union College*

Designing a micromouse PCB requires many considerations. For one, the micromouse will be subjected to a substantial number of collisions. The designer needs to make sure that the PCB will survive these crashes. If the PCB will serve as a support for the robot then it must also be large enough to carry all the components and be strong enough to handle their weight. However, it must also remain as small as possible to help with maneuverability.

Component placement is also a very important detail that cannot be overlooked. The placement of some components can impact functionality directly. Sensors and motors for example, need to be placed optimally to make sure they serve their purpose. Weight distribution is also another factor of component placement. The designer must be sure the micromouse is well balanced or one side of the mouse will be raised higher than the other, causing the wheels to skip for example.

There are also electrical considerations to be made. In relation to component placement, the designer must also consider the connections made between the devices mounted on the PCB.

Components need to be mounted in close proximity to simplify routing traces. The traces of the PCB (*figure 3*), which are basically etched wires on the board, are limited in their current carrying capacity. Depending on the current requirements of a mounted device, the trace must be a certain width to supply that demand. The space between conductors mounted on the printed board must also be considered. Positioning them too close to each other would cause induction or another form of interference.

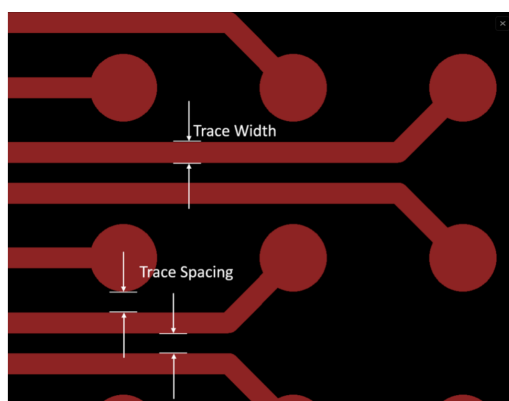


Figure 3. *Trace Width representation*

Lastly, any good design would have to consider some standards. The printed board industry widely recognizes the IPC (Institute of Printed Circuits) as the leading publisher of printed board standards[5]. Using IPC standards ensures the PCB design is up to date with current industry practices that produce quality products. Understanding these standards also helps reduce cost since most manufacturers charge extra money when requested to make a non-standard PCB, since making it would require a change in how their machines operate.

3.0 Design Specifications, Constraints and Standards

During the course of the fall term, the UCRC has purchased and tested components that will be mounted on the micromouse. *Table 1* shows the relevant component features that will need consideration when designing the printed circuit board. These considerations function to

constrain the PCB design and serve as the basis for constructing the PCB specifications. This was done because not all features of the components placed on the PCB have a direct impact on the PCB design. For example, the microcontroller used in this design is the Adafruit Feather nRF52840 Sense. Two of the main reasons it was chosen over other microcontrollers was because of its bluetooth communication capability and that it can be programmed using Circuitpython, which is the programming language of choice for the UCRC. However useful and important these two features are to the micromouse design overall, they have no direct bearing on the PCB design specifically — the bluetooth module here does not directly determine how the microcontroller is wired or positioned on the PCB since the module can be used wirelessly. So when deciding which features of a the components populating the PCB would impact the PCB design directly, these features were determined to be most important to consider:

1. Size and shape

The component dimensions and its shape will determine how much space it would occupy on the PCB. The larger the component, the less options there are for its placement on the PCB. The size also determines the quantity of that component that could be placed on the board, as well as what other components are possible to place alongside it. Its shape has a similar impact. For example, the microcontroller used in the PCB is rectangular in shape, which limits the space available next to its shorter sides but will have more space next to its longer sides.

2. Weight

The weight of the components will determine how durable the PCB needs to be to carry all of them. Heavier components might require a thicker PCB to house all components or might even require switching over to a more durable PCB substrate.

3. Current draw

Each component has a current requirement unique from other components on the board. However, there is not a universal trace width that can cater to each unique requirement. Although a choice could be made to only use traces capable of withstanding the largest amount of current drawn on the board, using wider traces than necessary for all connections would occupy more space on the PCB than necessary, which will in turn make proper conductor spacing more difficult. (see standards section)

4. Pin Count

The number of pins on a component would determine the number of vias and pads on the PCB as well as how many routes are required to connect them to other parts of the board. The more pins that have to be routed, the more complicated the design becomes.

5. Component package type

This refers to the use of either through-hole technology (THT) or surface-mount technology(SMT) for components. The design process and complexity of assembling the micromouse PCB drastically changes depending on the device package. (see design alternatives section)

Component	Size	Weight	Maximum Current Drawn	Number of Pins Used	Package Type
Adafruit Feather nRF52840 Sense + DC Motor + Stepper FeatherWing	51mm x 23mm	10.6g	Not needed/ depends on connected devices	21	THT
150:1 Micro Metal Gearmotor HP 6V with Extended Motor	2.6 cm x 1.19 cm	9.5g	1.6A	Uses encoder pins since they are attached	THT

Shaft					
Magnetic Encoder Pair Kit with Top-Entry Connector for Micro Metal Gearmotors	Not needed	1.5g	Same as motor	6	THT
Sharp GP2Y0A51SKoF Analog Distance Sensor	13.2mm x 14.2mm	2.7g	0.12 A	3/sensor=18	THT
Snap-Action Switch with 16.3mm Roller Lever	Not needed/ mounted horizontally	50g	5A	2/bumper=4	THT
Blomky 11V, 1500mA lipo battery	7.3 cm x 3.5 cm	108g	Total + 10% = 7.22A	Barrel jack will have two	THT

Table 1. *PCB Component Features*

3.1 Special considerations

Other considerations made apply to only a specific set of components but are necessary regardless:

1. Lead size and shape

Although most components can fit within a standard size via, the barrel jack leads and the bumper switch leads both have unique flat leads that require their own special vias.

2. Freedom of orientation

Aside from the obvious limitations of component orientation that stems from the size limitation of the PCB, certain components like the bumpers and the IR sensors have their function tied to their placement on the board. (see PCB layout section)

3. Heat dissipation

Since the micromouse dissipates very little heat overall this is a low priority consideration but must still be made.

4. Choice of Manufacturer

The choice of manufacturers impacts the type and quality of the PCB, the price, and the turnaround time required to produce the PCB. For this project the most important factors considered are the cost and the turnaround time. These factors will determine impact the schedule of production as well as the budget required for the project.

Using all of these considerations, the PCB specifications were constructed. (*table 2*)

Category	Specification	Justification
Area	10cm x7cm	This is within the 25 cm x 25 cm allowed by the micromouse competition and is also small enough to allow for decent maneuverability within a maze block.
Type	Rigid	This application requires a PCB that could withstand low impact
Quantity	5 boards each (primary and secondary PCB)	With 5 boards even if there are defective boards within this amount there are plenty of spares
Carrying Capacity	200 g	This is the total amount of weight the PCB is expected to carry including a 17g headroom. (calculated on table 4)
Layers	Multilayer (>1)	Due to the high circuit density of this project and the limited space on the PCB, routing would be very difficult with a single layer.

thickness	1.6 mm	This is a standard PCB thickness, it should be able to carry around 200g of weight.
Conductor Spacing	0.05 mm to 0.13 mm	Based on IPC standards a board of this type should include (see standards for more detail)
Cost	<\$50	PCB prices go higher the more customized they are. Standard PCB builds like the one for this project should not cost more than this.

Table 2. *PCB specifications*

3.2 Standards

The standards and guidelines used for designing the micromouse PCB were based on IPC-2221A. The sections that are directly applied to the project are presented below with their details.

❖ Conductor spacing [6]

- *Table 6-1* in *section 6* [appendix] lists the minimum space allowable between two conductors on a PCB based on the voltage difference between them. *Section 6.3* on ‘Electrical Clearance’ recommends maximizing space between conductors whenever possible but to avoid spacing smaller than listed on *table 6-1*. This standard is used to determine the distance between traces, vias and pads, as well as distance between conductive components populating the PCB.
- Standard application - The battery powering the micromouse is rated at 11V which lies between 0 to 15V range listed on *table 6-1*. This corresponds to a minimum space of 0.05mm for internal conductors (category B1) like traces, and 0.13mm between uncoated conductive components in the final PCB (category A6). According to the JLCPCB (chosen manufacturer - see design alternatives) website, their manufacturing process includes applying epoxy coating (a form of conformal coating for protection) on their PCBs. This puts the micromouse PCB in

category B4 under *table 6-1* where the recommended minimum conductor spacing between pads/vias on the PCBs is 0.05mm.

❖ Trace width calculation

- *Section 6.2* on ‘conductive material requirements’ contains *equation 1*. Which is used to calculate cross sectional area of traces on the PCB. The area is then used in *equation 2* to calculate the width of traces. These calculations are necessary because trace width limits current flow through a trace. For this project an online trace width calculator[9] was used to compute trace widths required by each mounted component.(the calculator also uses this IPC standard - see design section for details)

$$I = k\Delta T^{0.44}A^{0.725}$$

Equation 1. Current draw equation, where:

I = Current drawn in amperes

K = 0.024 for inner layers

K= 0.048 for outer layers

A = cross sectional area and

ΔT = Rise in temperature

$$W = A/[T(1.678)]$$

Equation 2. Trace width equation, where:

W = trace width

T = thickness of board

A = cross sectional area

❖ test points

- *Section 3.5.6.5* refers to the use of vias and pads as test points for post fabrication as long as they are not coated with insulating materials. Only through hole components are used for this project so component leads and vias will be probed directly to measure voltages and currents. (see testing section)

In addition, *section 10.1.3* on recommended routing angles and *section 4.3* On laminate & adhesive selection for structural strength were considered. However, neither of these apply to this project since the use of strict 45 degree routing angles, instead of 90 degrees, only applies to high speed applications (>10GHz), and the chosen manufacturer for the micromouse PCB —JLCPCB— does not allow selection of custom laminates or adhesives. The latter would also increase total cost.

4.0 Design Alternatives

This section covers the design approaches considered to implement the specifications mentioned in the previous section. Each design approach (represented in *table 3*) is accompanied by an explanation of the choices available and justification for the approach chosen. Note that approaches regarding component selection for the micromouse were made in consultation with the UCRC.

	Specification	Approach
#1	Materials	Fr - 4
		Aluminum
#2	Layers	2 layers
		4 layers
#3	Design Software	Ki CAD
		EasyEda
		Eagle CAD
#4	Motor driver type	Motor driver board
		Adafruit motor controller shield
#5	Battery	Stacked smaller batteries
		Large single battery
#6	construction	THT

		SMT
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Table 3. *Summary of Design Approaches*

4.1 Choice of materials

The material choice for the PCB impacts the strength of the PCB, its heat dissipation, and whether or not it can be used for high frequency applications. Aluminum and fr 4 were the final materials considered because they were consistently available for production from manufacturers. Other materials such as rogers, copper and HDI were less widely available and were expensive when compared to aluminum and fr-4. As stated before, the micromouse components do not generate sufficient heat to cause concern. Even if they did, fr-4 is made to withstand around 140 degrees celsius and aluminum can handle temperatures higher than that. When comparing the prices of these two, fr-4 was slightly more expensive (exact dollar amount changes between manufacturers but was lower than \$5)[7]. However, aluminum was only available single layered from JLCPCB, and was only available single or double layered from pcbway. Because of these limitations, fr-4 was chosen.

4.2 Layers

Initially it was assumed that 2 layers would be enough to complete the necessary routing. The first layer would be used to make the IO connections while the second layer could be reserved for the power and ground connections. However, early experimentation showed that because of the positioning of the sensors, motors and microcontroller, it was not possible to make all IO/ground and power connections in a single layer. Switching locations for components in the tight space that is available was too much of a downside so the layer number had to be doubled (PCB layers only come in even numbers so 3 layers was not an option). The cost added as a result is an extra \$4.

4.3 Design Software

The PCB design software is the tool used to create the printed board design. Most PCB design softwares have a schematic creation tool that is first used to make the circuit before it is converted into a PCB design. Once the design is completed, a gerber file can be generated. Manufacturers require gerber files to put a PCB design into production.

A list of pros and cons (see *table 4*) was used to determine which software to use for designing the PCB. The list was narrowed down to the three options that were most popular. Easyeda was eventually chosen because of its simple and intuitive UI, my lack of experience with the other softwares, and because it is directly linked to JLCPCB, which makes ordering the PCB easier.

Software	Pros	Cons
Easyeda	➤ Have prior experience making schematics on it ➤ Simple to use ➤ Large library ➤ Free ➤ Can directly order from JLCPCB	➤ Reviews complain about slow loading
KiCad	➤ Community updates libraries and symbols ➤ Free ➤ Flexible trace routing	➤ No prior experience with software ➤ A bit overwhelming (requires more practice) ➤ Must export gerber
Eagle CAD	➤ Free version is good enough ➤ Very popular ➤ Community support	➤ Paid for full features ➤ Must export gerber file (slight inconvenience)

Table 4. *PCB design software pros and cons*

4.4 Motor driver and battery implementation

The sparkfun motor driver breakout board used last year was replaced with a stackable motor driver from adafruit because it saved space on the PCB. The tradeoff was the added height of the microcontroller. In addition, a complementary decision was made by the UCRC to make the buttons on the microcontroller accessible by stacking the microcontroller on top of the motor driver instead. Since the motor driver's terminal blocks do not fit underneath the microcontroller, traces will be used to connect them to the motors instead.

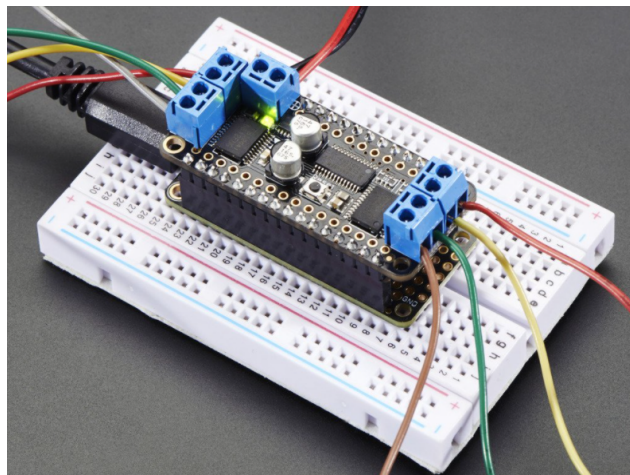


Figure 4. Featherwing add-on stacked on top of an adafruit microcontroller

There were two ways of implementing the battery on the PCB. Either a platform could be placed to house the battery (since it would not fit on the PCB directly), or to connect two smaller separate batteries in series to supply the equivalent voltage. This decision has not been made yet but in the meantime the team will use a platform to mount the larger battery.

4.5 PCB construction (pros and cons of THT vs SMT construction)

With through-hole PCB construction, the component leads go all the way through the PCB vias which obstructs traces made through all layers of the PCB. Surface-mounts however, come in much smaller sizes and can have components attached to both sides of a PCB area because they don't use through-hole vias that connect both sides. The biggest downsides to SMT devices is the

difficulty involved with soldering them. SMT devices can get so small that some of them might require the use of a hotplate and microscope to properly solder them onto PCBs. Soldering the smallest PCBs is usually left to the manufacturer's discretion. Lastly they don't offer breakout boards that simplify interfacing with complex components like the microcontroller and motor driver. For this reason a decision was made to use through-hole components exclusively for this project.

4.6 Manufacturing

JLCPCB was chosen for manufacturing the project PCB. Other companies that were contacted to send PCB files required complete gerber files to send back any quotes. The ones that did not require a gerber file (Nextpcb and Pcbway) offered much higher prices (around \$23 - \$25) for similar PCBs. This was simply unnecessary since JLCPCB already provided a short turnaround time of around a week and offered low prices (\$2 to \$15 for 10 boards). Their company also owns the easyeda software which means the complete design could be sent straight to them. It is also important to mention that these prices are only the costs for the boards themselves. Since all three of these companies manufacture in China, the shipping costs are also quite high (\$21 for JLCPCB for example). To avoid these high shipping costs local manufacturers were also contacted. Royal Circuits, for example, were sent a rough gerber file for a quote which came to \$699. JLCPCB was indeed the cheapest and fastest option available.

5.0 Final Design and Implementation

This section breaks down the entire design and implementation process starting from the top level to the schematic and ends with the detailed descriptions on the bottom level design. This includes the component layout, routing decisions and calculations made and design improvements made along the way.

5.1 Top Level Design

In its simplest form the micromouse has four key components. These are the microcontroller, IR sensors, motor driver and two batteries. Each of these components will be mounted on the PCB either directly or indirectly so an understanding of how they are connected and how they interact with each other is imperative for the PCB design. As shown in *figure 6*, the microcontroller serves as a central hub where it can interact with all the other components. It is responsible for taking in the inputs from the IR sensors and the motor encoders to gauge the robot's speed and location within the maze. Specifically, there are six IR sensors that send the microcontroller data on how far the nearest obstacle is. The encoders' function is to send data on the speed and direction of the motors they are connected to. Based on its prior programming, the microcontroller will then use these inputs to calculate and send a set of outputs to the motor driver. These outputs will determine the new direction and speed of the motors. The motor driver's function is to provide sufficient current to the motor. There are also three bumpers that will alert the microcontroller to collisions from the front, left and right sides. Lastly, there are two batteries mounted on the micromouse. These will provide power to the microcontroller and motor controller separately.

It is important to mention that the battery driving the motors (through the featherwing motor controller) is smaller than the 6V requirement listed in the device datasheet. However, after experimenting with a lower voltage lipo (3.7V), it has been determined that the only downside to the smaller battery is a lower maximum speed.

This is an acceptable change since the UCRC plans to run the micromouse slowly. Lower speeds will give the micromouse more time to react to the changes in its environment. Leading to less problems if the sensors encounter an error.

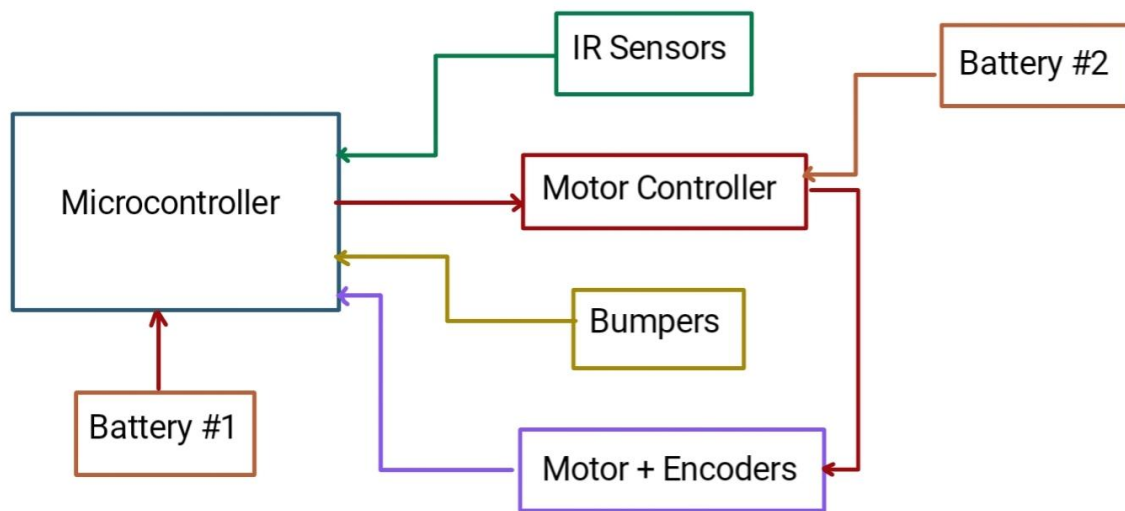


Figure 5.Block diagram of Micromouse Operation

5.2 Micromouse Schematic Overview

The components populating the micromouse PCB are connected as shown on *figure 7*. Since the stacked motor controller and microcontroller share most of their pins, the schematic only displays the motor controller whose pins are directly connected to the PCB. The figure shows the micromouse uses a total of 13 IO pins on the microcontroller. Only six of these pins have ADC (analog to digital converter) peripherals and are all connected to the IR sensors. Aside from the signal pins, each IR sensor also has one power and one ground connection made with the microcontroller's GND and 3V pins respectively.

The remaining pins are split between the bumpers and the motor encoders. The bumpers use three signal pins —pins 5 ,6 and 9—and the encoders use pins 10 to 13. All 7 of these pins have PWM peripherals but only the encoders will use them in their operation.

The motor controller power supply uses a barrel jack port which is directly soldered onto the PCB. The microcontroller power supply however, is not. The lipo battery is instead connected directly to the JST port on the microcontroller. Finally, the motor input pins on the encoder will be connected to the H-bridge pins on the motor controller (M1 & M2). These pins have to be

connected in pairs to allow the motors to switch directions. Since the micromouse has two motors, that means four connections are made in total.

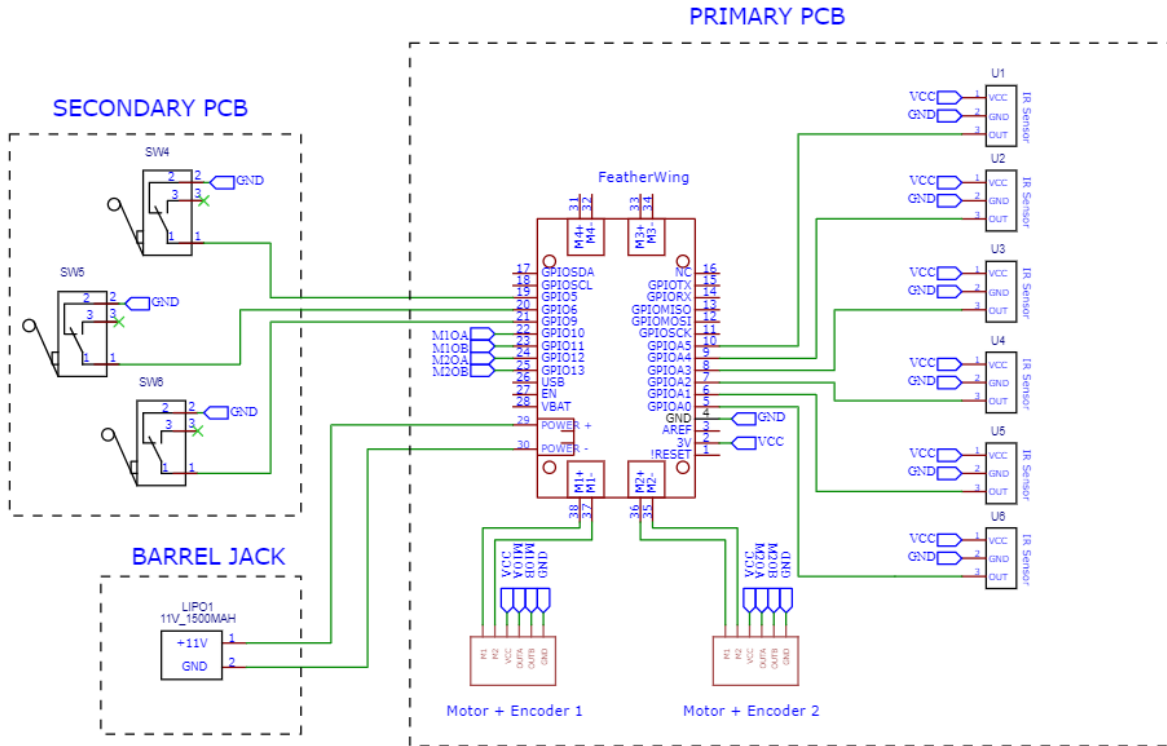


Figure 6. Schematic of PCB Micromouse (original)

5.3 PCB Design Overview

The design process of the micromouse PCB was iterative. As a result, multiple prototypes were manufactured and tested. Some of these designs failed to meet expectations or were subject to design changes made by the robotics team. As of this moment there are three iterations for the micromouse PCB. The limitations of each design, the tests conducted to reveal these limitations, as well as the subsequent improvements made, are all documented in the sections below.

5.3.1 First Iteration PCB

This design includes two separate PCBs connected to each other. The primary PCB will house the microcontroller, six IR sensors and two motors with attached encoders. The secondary PCB will only house the three bumper switches. Lastly, a barrel jack will be used to connect the motor controller battery to the PCB. The battery itself is mounted on a platform attached to the primary PCB. The microcontroller battery is also located on this platform. The two batteries are stacked and fastened with velcro strips.

5.3.1 a) Layout

IR sensors - These are mounted around the front of the PCB. The exact locations are shown with silkscreen markings on *figure 10* in yellow. These sensors do not come with rigid leads or headers, nor do they have mounting stands. Instead they come with JST-ZH type cable ports that are made to connect to flexible JST cables. For installation on a PCB the manufacturer recommends either soldering them sideways to the JST port as shown on *figure 8* or gluing them upright on the PCB to keep them stable while mounted [8]. The latter option was chosen to save space.



Figure 7. *Suggested soldering position for the GP2Y0A51SK0F analog distance sensor*

The silkscreen markings for the IR sensors are accompanied by 3 pads that connect to power, ground, and the corresponding microcontroller pin. The ‘power’ and ‘ground’ connections are marked to avoid confusion. Once these are manufactured, the JST cables will be soldered directly onto these pads.

The GP2Y0A51SK0F distance sensors have a total range of 2cm to 15cm. Outside of this range they send out unreliable data. To avoid this, an attempt was made to place the sensors 2cm behind the edge of the board they would be pointing towards. However, because of the limited footprint of the PCB, and its rounded front design, this was not achievable. The furthest back the sensors could be placed without overlapping with other mounted components or crowding traces was around 1.2 cm. Later sections will discuss improvements made in this area. But for this iteration, the protruding bumpers were assumed to offer a partial solution, since they would likely be triggered if any obstacle entered this range.

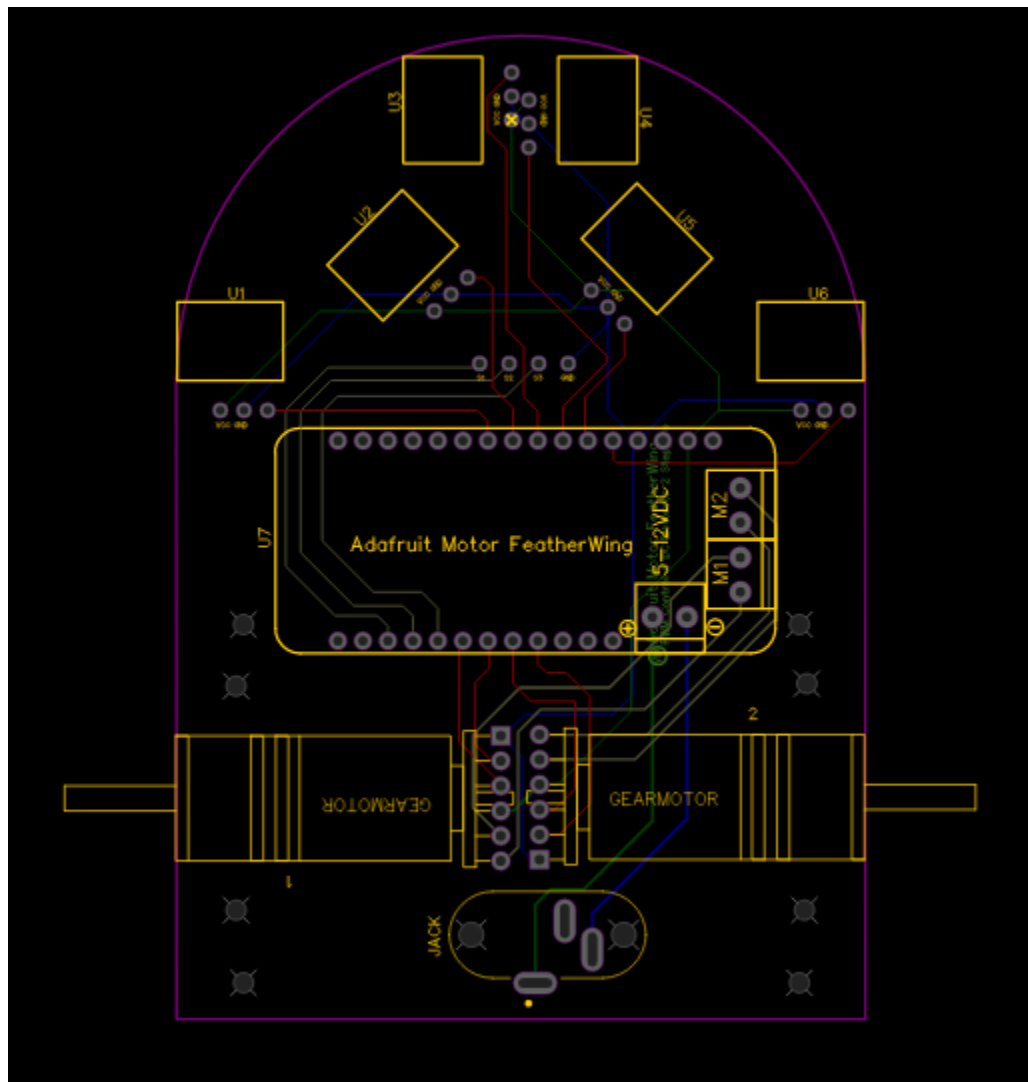


Figure 8. *Primary PCB Design*

The placement of the sensors on the board is as follows: The front direction sensors are placed on the left and right edges of the PCB (marked 'U1' and 'U6' on *figure 10*, respectively). The right and left direction sensors are placed at the front of the PCB ('U3' and 'U4'). The last two sensors are directed 45 degrees to the left and right of the PCB (marked 'U2' and 'U5'). There is adequate space in between the sensors to avoid interference.

Motors - These will be placed 16mm from the back of the PCB. Each motor has a screw hole on both sides where a wheel mount will be attached. At this point in the design process the UCRC had not yet constructed the wheel mount, so the location of the mounts was estimated. Ideally, the motors would be placed closer to the center of the PCB to balance the micromouse; however, there needs to be enough space for routing between the microcontroller and the motors, as well as adequate space for screw holes for mounting the battery platform. To compensate for the weight imbalance the battery platform will be mounted closer to the back of the board.

Motor driver & Microcontroller - The microcontroller is placed in the middle of the board to minimize the length of traces and to provide easier routing access to other components mounted on the PCB. For example, the analog pins that will be connected to the IR sensors are facing the front of the PCB (where the sensors are located), and the PWM pins connected to the motor encoders are facing the back of the PCB (where the motor encoders are located). Ideally, to continue this optimal setup the bumper switch pins would be located at the front where they are more easily accessible. However, all the microcontroller pins facing the bumpers were already in use.

*Note - The motor driver footprint was edited to remove the M3 & M4 pins since they are not used and would only obstruct traces.

Secondary PCB - Between the IR sensors and the microcontroller, there is a row of 4 pads where the secondary PCB will be attached [see *figure 10*]. Left to right, there are three pads for the bumper switch outputs, followed by the ‘ground’ pad. This position was preferred to keep most of the micromouse weight as close to the motors as possible without interfering with other devices on the board.

This PCB has the same width as the primary PCB but is much shorter (40.2mm). Soldered headers attach the two PCBs together at the location of the 4 pads mentioned above. Note the secondary PCB is attached to the bottom side of the primary to prevent obstructing the IR sensors. The solder joint is the only mechanical connection between the two PCBs.

Three switches are installed horizontally into the U shaped pads at the front, left and right of the PCB. This positioning allows the micromouse to detect collisions where they are most likely to occur. Having more bumpers would improve collision detection but would also add too much weight to the front of the PCB. The inverted T shape of the board shown on *figure 11* was preferred to reduce unnecessary weight which would otherwise strain the solder joint greatly.

Unlike the primary PCB this board only has two layers. One layer is reserved for the ground connections while the other is reserved for the switch outputs. Despite its low circuit density, without the two layers it would be very difficult to make all necessary connections on this PCB without crossing wires. A 3D render of the secondary PCB is shown below.

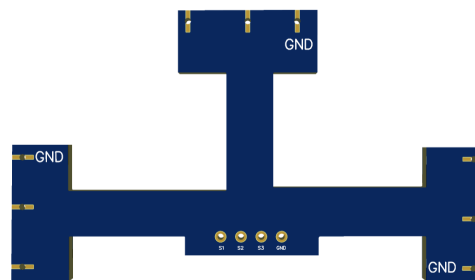


Figure 9. *Premanufacture 3D Rendering of First Iteration PCB design*

5.4.2 b) Routing

The routing will be broken down between all 4 layers. The ground and power traces will be located on separate layers while the other two layers will be alternated between the remaining connections. Trace width calculations were made using data on current draw from *table 5* and *equations 1 & 2* from the IPC standards. The resulting minimum trace widths that will be required are shown on *table 5* Below. However, since manufacturers are limited in how small they can make traces, any trace smaller than 0.2 mm will be defaulted to 0.2 mm which provides 5.95 A.

Component	Current Draw (A)	Trace Width (mm)
Motors	1.6	0.0327 (default)
Sensors	0.12	0.000918 (default)
Bumpers	0.010 (logic pin max output)	0.00117 (default)
Battery	7.22	2.61

Table 5. *PCB Components Current Requirements and Calculated Trace Widths*

Progress

The primary PCB was manufactured during the winter break and delivered at the start of the term. No components were soldered onto it but the initial continuity tests showed the design was accurate to the schematic sent to the manufacturer. *Figure 12* shows the manufactured board.

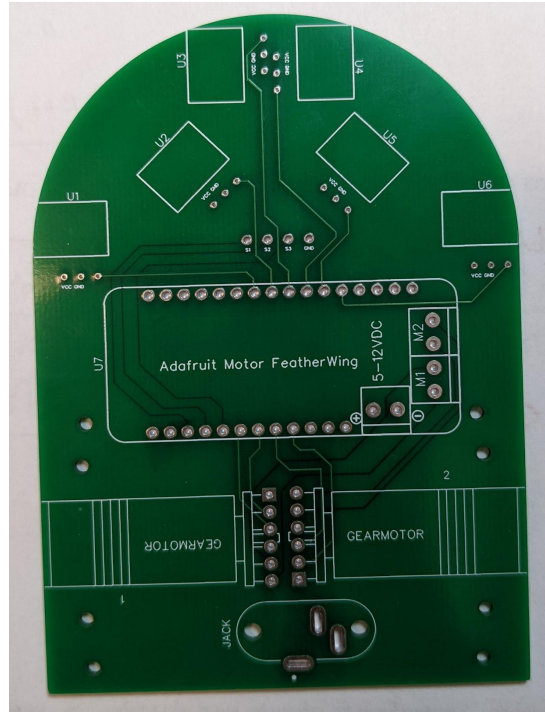


Figure 10. *Manufactured First Iteration PCB*

The secondary PCB did not reach the manufacturing stage. The main issue that delayed its production was that the U shaped pads required to mount the bumper switches horizontally were not standard and so had to be constructed manually. One solution attempted was to have the board outline wrap around the switch pins instead of using U shaped pads that were unavailable. However, this solution was never implemented since the robotics team made major design changes at the start of the term that overhauled the current PCB design.

5.3.2 Second Iteration PCB

After consulting with the robotics team, several substantial changes were made to the design of the micromouse PCB. First, the motors were moved to the center of the PCB. Ball casters were added to improve balance. The secondary PCB was removed entirely from the design and lastly, ground and power planes were introduced. Details are discussed below.

1. Moved motors to the center of PCB

The biggest design change in the second iteration was moving two motors from the back to the middle of the PCB. The microcontroller was moved to the back in its place. This change will move the micromouse axis of rotation to the middle of the robot where it will be easier to control. Think of the difficulty of rotating a shopping cart from the back versus rotating it along the middle. In addition, rotation with the previous setup also would have required swinging the entire length of the robot. Which would require more space within the maze. This iteration would improve upon that as well. A helpful illustration of that idea is shown below.

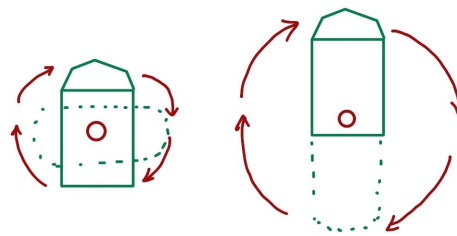


Figure 11. *Illustration of difference in radii due to change in axis of rotation*

I have mentioned in the motor layout section that this was the ideal position for motors to balance the micromouse but that doing so would restrict the microcontroller's access to the front components and restrict routing paths. The last two changes help mitigate these issues.

2. Addition of ball casters

Initially, the micromouse was supposed to have two wheels attached to each motor to balance it out. Other members of the robotics team were working to design a motor mount to make this possible. However, it was a complicated solution and easier alternatives were available. So it was decided that ball casters would be attached to the front and back of the micromouse instead. To make space for the ball caster screws, the barrel jack was moved to the bottom left and the IR sensors were moved slightly downwards.

3. Removal of secondary PCB

Since the IR sensors were giving accurate readings on breadboard tests it was decided that they would be relied upon to detect collisions as well. They won't alert the micromouse of a crash like

the bumper switches would, but they can detect when an obstacle gets too close. Removing the PCB also meant there was more space available for the ball casters at the front and for the IR sensors to be moved back. As a result the 45° IR sensors were moved 1.8 cm away from the edge of the primary PCB, which is close enough to the target 2 cm which will give more reliable data at close range.

4. Addition of ground and power planes

The final change was made by myself to reduce the number of traces used in the design. By removing the solder mask isolating the common ground and power pads on the PCB, an entire layer on the board can be used as a conductor. The rest of the pads on the board remain isolated to prevent shorts. Because this method drastically reduces the number of traces used, the design was changed to a two layered (double sided) PCB instead of the 4 layered one used in the first iteration.

The top layer of the PCB was converted to the power plane while the bottom was converted to the ground plane. The traces for the remaining connections were split between these two planes.

Figure 14 has a helpful illustration that summarizes the changes made for this iteration. The manufactured board is shown alongside it for comparison.

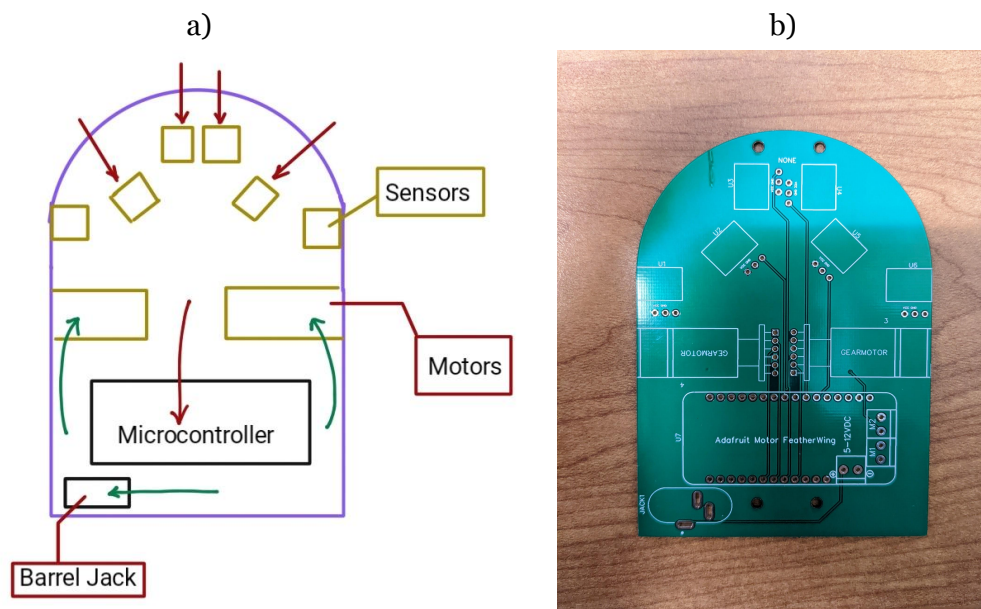


Figure 12. *a) Illustration of changes made in the second iteration. b) Manufactured PCB*

Progress

Although this iteration showed promise, an incorrect gerber file was sent to the manufacturer. The version that was sent included the major changes made from the first iteration but contained fatal defects that kept it from being fully assembled. Luckily most of these defects were relegated to the left motor and so partial assembly and testing was still conducted. Here is a shortlist of the issues encountered with this version:

- Missing connection between left motor and microcontroller
- Two encoder pins were shorted
- Adequate space was not reserved for the motor mounts. Which led to an overlap between the front IR sensor pads and the screw holes for the motor mounts.
- Incomplete silkscreen markings

5.3.3 Third Iteration PCB

The third and final iteration of the PCB micromouse resolves the issues listed in the previous section and adds a few extra features. Notably, the connection issues have been removed and the design is now accurate to the schematic. In addition, the front IR sensor pads were moved to the side to make space for the motor mount screws.

This version also adds more helpful silkscreen markings into the design. The IR sensors now have extra labels like 'L' & 'R' to show the direction they should be facing. This will make testing less confusing for members of the programming team who are less familiar with the design. The IR sensor logic pins on the microcontroller are also labeled so they'll be easier to find when doing continuity tests or debugging. In addition to practical upgrades, aesthetic changes have been added as well. There is now a Union College logo on the bottom right with the designer and team name written on it.

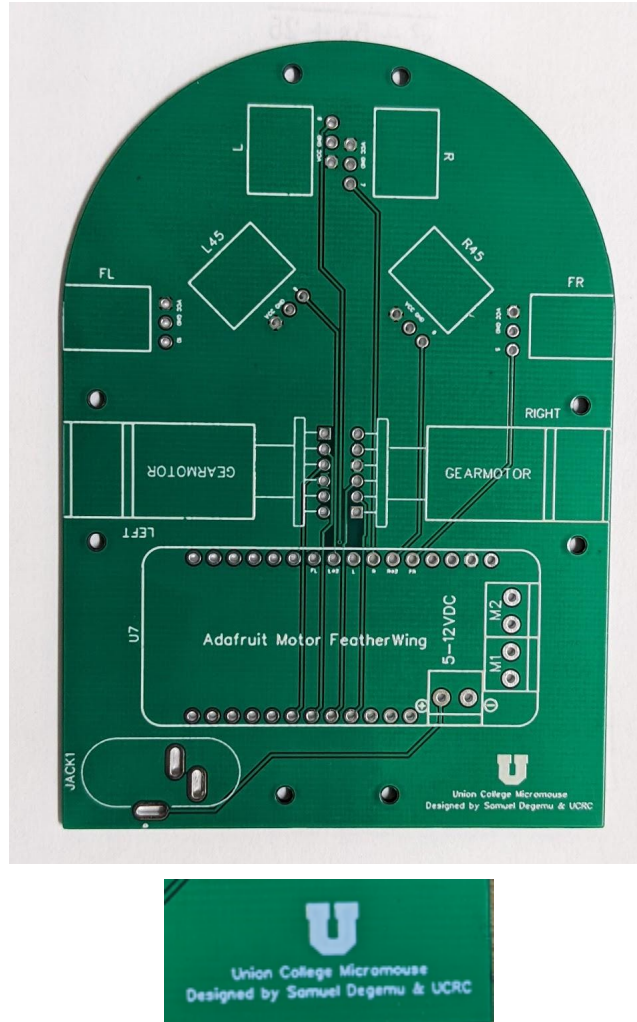


Figure 13. *Final PCB (with logo)*

Aside from the changes in the actual PCB design, the battery used for the micromouse was also replaced. Initially a 12V, 2200mAh lipo battery was supposed to power the motors(via the barrel jack). This was switched to a much smaller and lighter 3.7V, 2200mAh battery that could also be used to power the microcontroller. According to its datasheet, the motors are designed to perform at 6V. This battery is also smaller than the 5V requirement for the motor controller. Nonetheless, experiments with the lower voltage lipo (3.7V) showed the motor controller functioned properly. it has been determined that the only downside to the smaller battery is a lower maximum speed. This is an acceptable change since the UCRC plans to run the

micromouse slowly. Lower speeds will give the micromouse more time to react to the changes in its environment. Leading to less problems if the sensors encounter an error.

5.4 Conclusion

The final iteration of the PCB fulfills the requirements for the overall micromouse design. The small size and centralized axis of rotation improves the maneuverability of the micromouse within the maze. From the electrical side, most of the external wiring has been removed from the design through the use of traces and conductor planes. Repositioning the IR sensors (the 45° sensors in particular) further back has improved the overall collision detection of the micromouse and has eliminated the need for bumpers. Lastly, the addition of ball casters has helped with balancing the micromouse while replacing the batteries has reduced its weight.

6.0 Final Results and Design Evaluation

This section will detail the tests conducted on the micromouse PCBs followed by the results obtained from these tests. Although there were multiple iterations of the PCB, only the final version was tested and assembled fully.

6.1 Test Procedure

Before the PCB is sent out to manufacturing, the design rule check (DRC) tool on easyeda is used to confirm the design follows the specifications during review. The tool allows setting design rules or parameters for trace width, conductor spacing and connection pairs to raise an alert when any of these rules are broken. Using this makes sure the design is delivered as expected. The DRC tool was not used to review the second iteration PCB which led to a flurry of mistakes. Once the PCB has been manufactured each board is visually inspected for defects. Scratched, bent, or otherwise damaged boards are discarded. The boards that pass inspection are tested for

continuity using a digital multimeter. These tests reveal internal errors such as unintended overlap between traces or incorrect connections.

Each connection on the PCB is cross referenced with the circuit schematic and tested individually. The copper pads on the PCB are used to access the connections on the board with a multimeter probe. After the PCB has been populated with components, the continuity tests are repeated by probing the solder joints. If any failures occur here the joints are resoldered. The checklist shown on *table 6* is used to record the results of the continuity test.

Components	Connections		Before & After Soldering		Components	Connections		Before & After Soldering	
MCU	10	Left IR	✓	✓	MCU	3.3V	Motor 1 encoder VCC	✓	✓
	9	45° Left IR	✓	✓			Motor 2 encoder VCC	✓	✓
	8	Center Left IR	✓	✓			IR sensor VCC x6	✓	✓
	7	Center Right IR	✓	✓		GND	Motor 1 encoder GND	✓	✓
	6	45° Right IR	✓	✓			Motor 2 encoder GND	✓	✓
	5	Right IR	✓	✓			IR sensor GND x6	✓	✓
	22	M1OA	✓	✓	Motor	M1 +	Motor 1 M1	✓	✓
	23	M1OB	✓	✓		M1 -	Motor 1 M2	✓	✓
	24	M2OA	✓	✓		M2 +	Motor 2 M1	✓	✓
	25	M2OB	✓	✓		M2 -	Motor 2 M2	✓	✓
Barrel Jack	1	+	✓	✓					
	3	-	✓	✓					

Table 6. Continuity Test Checklist Result of Final PCB

One of the boards that passed all prior tests is then drilled with a 6/65" sized drill bit in four points. At the end of the assembly process, four spacers are screwed into these points to mount the platform and the two batteries of the micromouse. The drill point locations are shown on *figure 16* below.

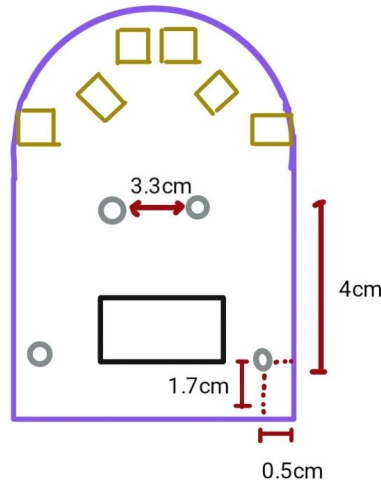


Figure 14. *Drill Points for Battery Spacers*

To begin component placement, sockets are placed into the featherwing pads and soldered. This prevents the extra step of desoldering the motor controller if a quick removal or replacement is required. Once the microcontroller is stacked on the featherwing and plugged into the sockets, the barrel jack is soldered onto the PCB. The 3.7V battery is connected to the jack and a multimeter is used to check if power is properly being supplied to the motor controller. If a problem occurs here, or at any point in the future and current measurements are required for troubleshooting, all the components can be partially disconnected from the board to break the circuit.

These components will then be soldered and tested in the following order:

- **IR sensors** - are installed one at a time, starting with front direction sensors, followed by the right and left direction sensors, then finally the 45 degree sensors. After the installations, the UCRC programming subteam will run test codes on each sensor to check for valid readings. At this point, since the continuity tests have shown the

connections to the sensors are correct, inaccurate readings will be assumed to be a result of a faulty sensor and a replacement sensor will be installed. If the errors persist, current and voltage measurements are made to check if the microcontroller is supplying adequate power to the sensors. If not, a design revision will be required.

- **Motors and encoders** - The motors are placed on their silkscreen markings and fastened to place with the motor mounts and screwed in. The JST cables of the encoders are run through the vias and soldered. The cables are shortened beforehand if necessary to prevent clutter. The wheels are connected one at a time. After installation, the programming subteam run their motor test codes and check to see if they are getting correct readings on motor rotations. Any errors that occur here are debugged through an oscilloscope. The exposed leads of the JHT cables run far enough underneath the board to allow attachment to oscilloscope alligator clips. These, and all other such leads are shortened after testing is complete. If the oscilloscope shows the encoder is responsive to motor rotations, the connection might have to be re-soldered.

Once all tests are completed successfully, the ball bearings and spacers are installed.

7 x 5cm rectangle pieces of plexiglass are then cut to use as a battery mount. The glass is then drilled to fit the spacers on the PCB. Plexiglass was chosen here because it is light and its transparency allows us to see the LED lights blinking on the microcontroller during testing. Below are a few pictures of the semi-complete micromouse.

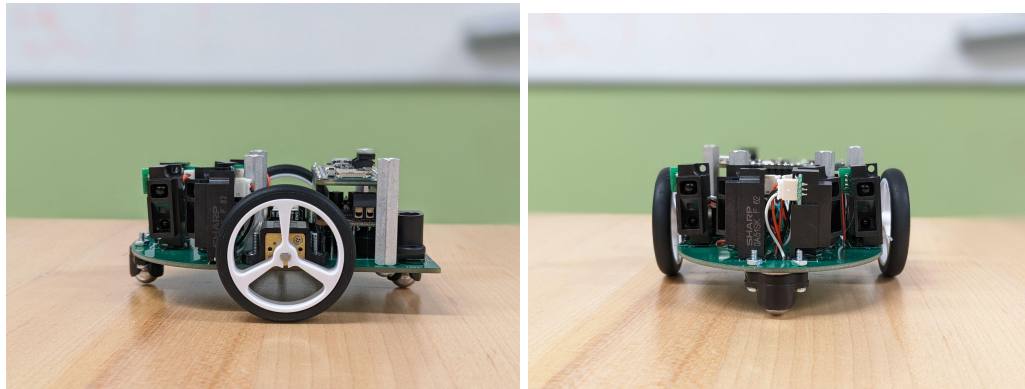
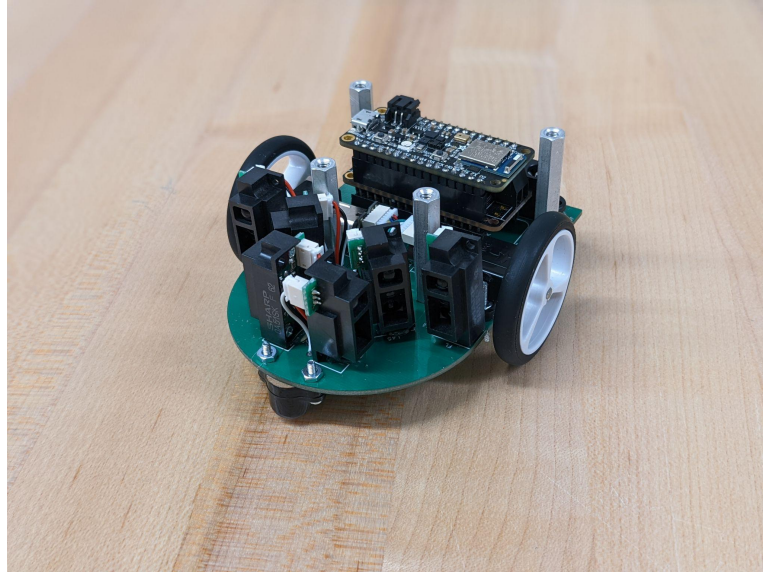


Figure 15. *Pictures of Assembled PCB micromouse (Missing batteries & plexiglass)*

6.2 Results

Continuity Tests

The continuity test checklist shown on *table 6* was used for all three PCB iterations. Tick marks (✓) were used to denote successful continuity tests while (x) was used to denote failed tests. Blank check boxes imply testing was not completed. The first iteration was not populated with any components so the entire right column of the checklist was left blank. Similarly the second iteration was only partially populated (Right motor and right sensor only) so it showed plenty of

blanks as well. The second iteration PCBs were the only ones with failed continuity tests before soldering. The other iterations had no connection errors post manufacture. The voltage measurements that were conducted after connecting the barrel jack were also successful. Both the 3.7V battery and the original 12V battery conducted well with the PCB.

Table 6 itself shows the results of the final iteration only but the results of the other iterations can be found in the appendix.

6.2.1 Test Codes

The purpose of running test codes on the motors and IR sensors is to show these components could send out proper signals without any interference. If both components are able to transfer accurate data to the microcontroller, then the PCB has achieved its objective. Assuming the manufacturers followed the instructions, this would also mean the electrical specifications (ie. trace widths & conductor space values) were met.

The first codes run on the micromouse were to test the IR sensors. These codes check the responsiveness of each IR sensor to the location of maze walls. To conduct this test two maze blocks were placed at the front and side of the micromouse. The micromouse is then connected to a computer via usb cable so the codes can be uploaded. The readings from the IR sensors are then plotted vs time on the plotter tool located in Mu code editor (This is the programming team's chosen code editor to run CircuitPython code on the micromouse). Each line on the plot represents a single IR sensor's readings. Although all the sensors are connected, only three of their readings are plotted at a time. This makes the data easier to read. Once the code is set up, the maze blocks are moved to stimulate a change in the sensor readings.

The test plots for the IR sensors on the right are shown below with brief descriptions of the situations that caused those results and what they mean. The results for the sensors on the left are placed in the appendix.

Front
45 Degree
Right

Note the color codes displayed here apply to all IR sensor plots shown below.

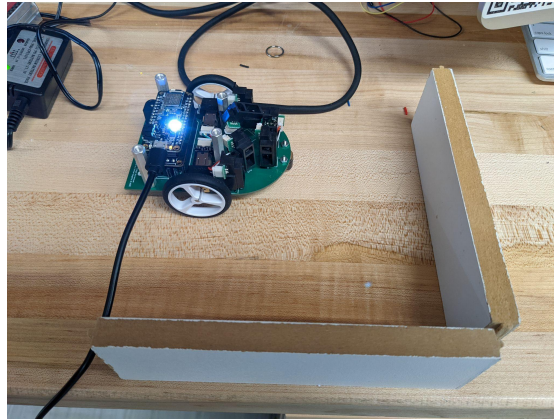
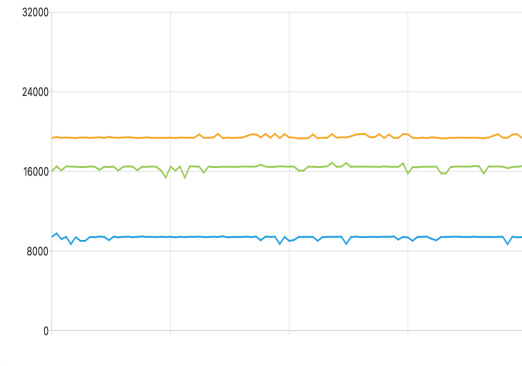


Figure 16. *IR sensor Plots - Scenario 1*

The plot above shows a stable reading across all sensors on the right side. This shows that the micromouse is sitting still. Higher analog values mean the walls are closer while lower values mean they are further away. The right sensor (orange) has the highest value in this case, which means the front wall is closer than the right wall.

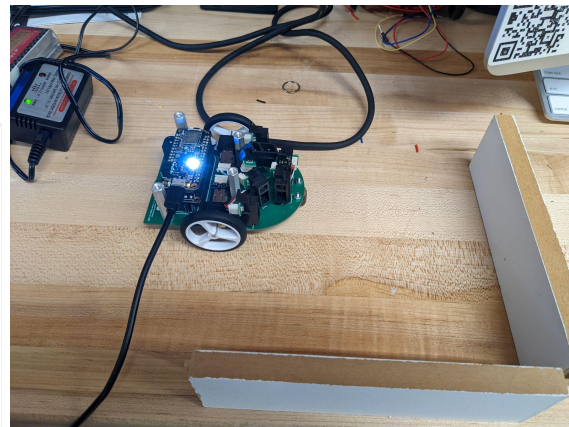
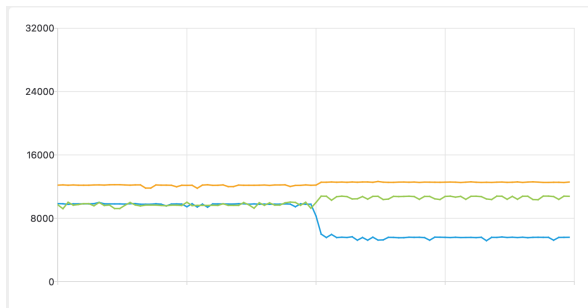


Figure 17. *IR sensor Plots - Scenario 2*

This shows a scenario where the front wall was moved further from the micromouse. The front sensor (blue) reading dips as a result. Readings from the other sensors remain the same with minor changes in the diagonal value. This is expected because the diagonal sensor (green) is

pointing towards the intersection of the two walls which is likely to move when either wall changes location.

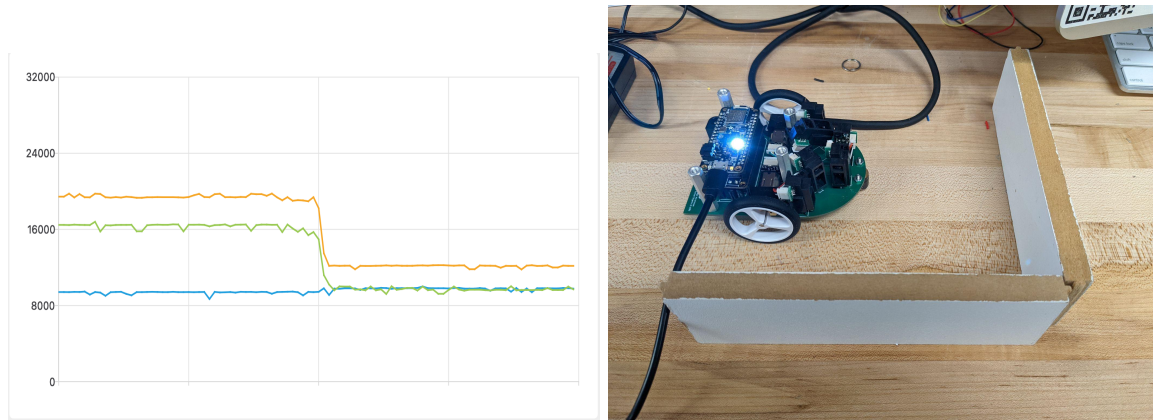


Figure 18. *IR sensor Plots - Scenario 3*

This is another example scenario showing dips in both right and diagonal sensor values. The front sensor (blue) in this case remains the same. This is a scenario where the right wall was moved further from the micromouse.

These tests were repeated with various combinations of positions. Including situations where the wall is against the side of the robot. The full range of the analog reading is about 200 to 45,000. The datasheet for these sensors suggest these analog readings would repeat between ranges 1.5 cm and lower which was not encountered with this design. (see plot in the appendix)

6.2.2 Motor Codes

The motor codes functioned similarly to the IR sensor codes. The encoder counts are captured on the serial monitor and plotted against time to show changes in speed. Since there are only two motors, both of their speeds are shown on the same graph.

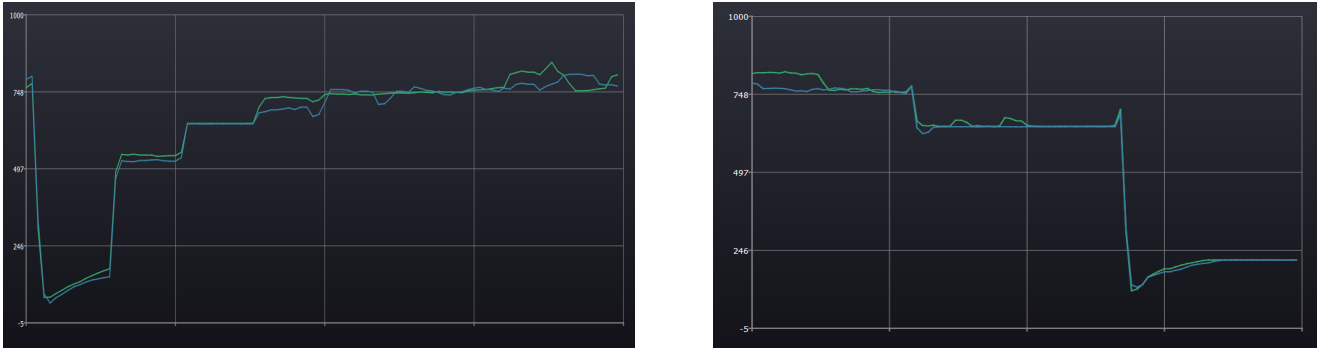


Figure 19. *Motor Encoder Readings for Left and Right Motors*

The first scenario shows an increase in speed from 0.1 to 0.8 speed values in intervals of 0.1 units. These values are scaled so that a speed of 0 means no movement while a speed of 1 runs the motors at their highest speed. The figure shows the left (blue) and right (green) motors run at similar speeds but are rarely ever the same. This is true in spite of them being programmed to run at the same speed. This is a common problem motors have. The graph on the right shows a decrease in speed from 0.6 to 0.1. The upper limit was lowered since the 0.8 speed from before was giving off very unstable results. It's important to mention that these specific results were obtained using the original 12V lipo battery. When switching to the smaller 3.7V battery, much more stable results were obtained.

6.3 Conclusion

From the results above it can be concluded that the design worked as intended. The DRC tool showed the first and final PCBs followed the schematic and had proper conductor spacing and trace width parameters. The motor encoders and IR sensors were also shown to be adequately responsive, which demonstrated the electrical specifications implemented work properly.

One way to improve on this design is to add logic buttons. Doing so could increase functionality. Since the bumper pins are no longer used, the button could be wired to one of those pins. The

reason a button was not considered in the first design is because it was assumed that the remappable button on top of the microcontroller would be easily accessible. However, after placing the plexiglass cover on the spacers, it has been very difficult to reach that point from the outside.

Finally, better testing results could have been obtained by using a logic analyser instead of just plotting the encoder counts. This would have provided better proof that the PCB met specifications since we would be able to look at the signals sent to the microcontroller and investigate the signals without having to guess.

7.0 Discussion, Conclusions and Recommendations

This section will summarize the project goals and results, make recommendations based on those results and will discuss lessons learned during the project. It will discuss shortly how the final design has met specifications. The recommendation section mainly discusses the future prospects of the project. It answers questions like What's next for the project? And What work can be done to improve the current design?

7.1 Summary

This project was undertaken to help the Union College Robotics Team make significant upgrades to their previous micromouse designs. Built on breadboards, these older designs had tall structures & cluttered and loose wiring. Converting the micromouse to a printed circuit board was assumed to fix all these issues. The design process for this project was iterative. Two versions of the PCB micromouse failed while a third one was successful.

Before manufacturing the PCB was put through design rule check which proved the correct conductor spacing and trace widths are being used for the design. Post manufacture and assembly the PCB micromouse performed satisfactorily in the tests conducted on it as well.

The continuity tests for the final PCB were all successful. And as shown on *figure 22*, compared to the other designs from the past, the physical goals of the micromouse were also met. The new design is much shorter and occupies less space than the older versions. It also includes a much lighter battery to help reduce its overall weight. Finally, the design is also structurally sound. Which will guarantee a long lifespan where future Robotics team members can use it to test new algorithms or even to test new feather microcontrollers. Since adafruit microcontrollers and shields share similar footprints, this one could easily unplugged from the sockets and replaced if necessary.

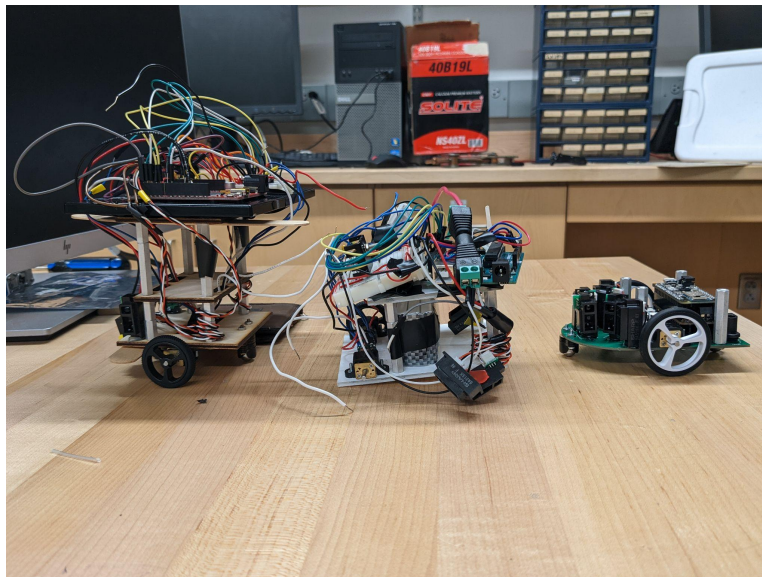


Figure 20. *Picture Comparing the PCB Micromouse to Older Designs*

7.2 Recommendations

As well as the micromouse PCB has performed so far it doesn't mean there are no improvements that could be made. As stated in the testing section conclusion, adding buttons would improve functionality greatly. Having a button to turn the micromouse ON and OFF is the first step to improving this design. The addition of an extra remappable button would also benefit the programming team who might implement a helpful new functionality.

The design file for this project will be left here so other students who wish to continue this project can inherit it and try out different design schemes. Adding sensors to the back of the micromouse for example could add an interesting new 'back-up to reset position' feature without having to worry about collisions.

Hopefully the robotics team could eventually move to using surface mount components to minimize the footprint of the PCB further. Alternatively, they could also keep the size as it is but populate the PCB with more components. Perhaps add the bumpers that were removed from the current design to increase functionality. Future students could also experiment with different shapes to use for PCB and keep iterating to find the best one.

7.3 Lessons Learned

This was my first time working with PCBs so there were many skills I had to learn. Aside from the more skills like routing PCB traces I also learned how to create custom schematic symbols and PCB footprints when I needed them.

I also learned how to use conductor planes to simplify my designs. This made soldering more difficult however, since heat from the soldering iron would dissipate into the rest of the board. Requiring that I press the soldering iron for a long time, potentially damaging pads with the heat. I learned that heating up the board before soldering helps alleviate this issue. In the topic of soldering I also learned that using lead based soldering for ground planes is a bad idea. The higher melting point of the lead solder meant I needed to heat up the soldering iron much more than normal.

8.0 References

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- [8] Sharp Electronics. (n.d.). *Pololu - sharp GP2Y0A51SKoF analog distance sensor 2-15cm*. Pololu Robotics & Electronics. Retrieved December 1, 2021, from <https://www.pololu.com/product/2450>.
- [9] *Advanced circuits*. Printed Circuit Board Trace Width Tool | Advanced Circuits. (n.d.). Retrieved March 18, 2022, from <https://www.4pcb.com/trace-width-calculator.html>

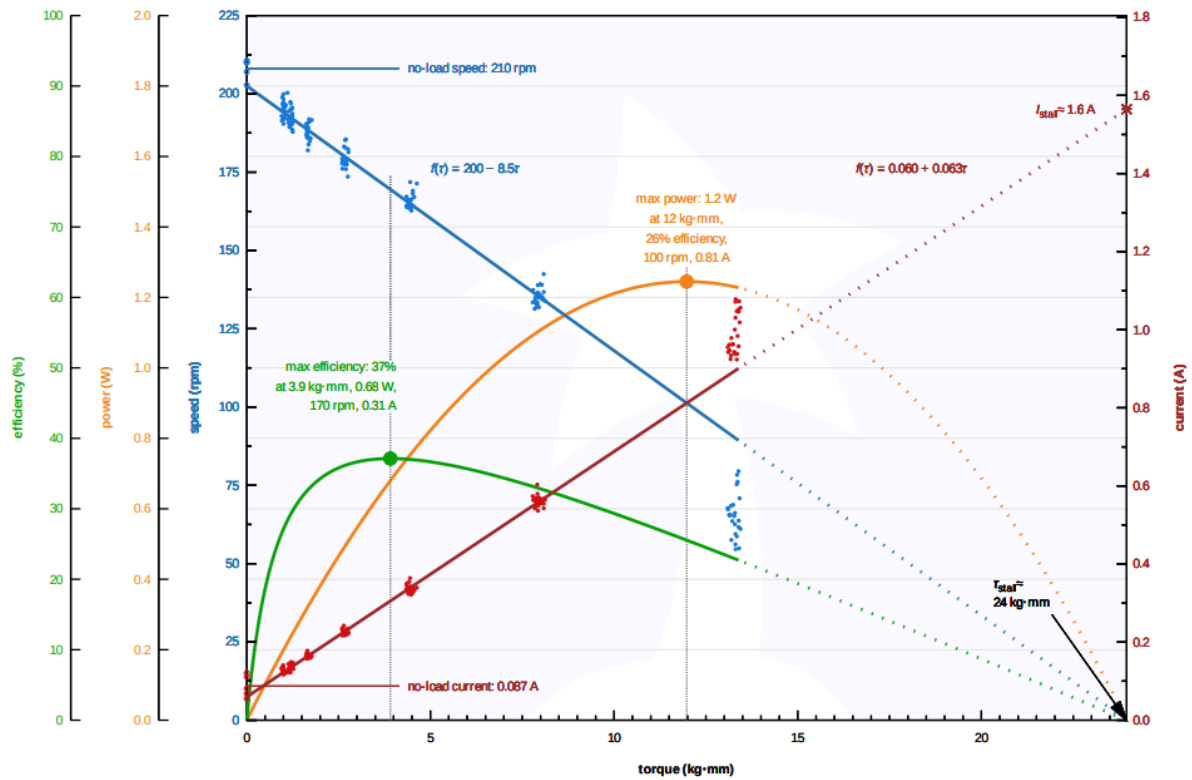
9.0 Appendix

Product reference links

- <https://www.pololu.com/product/1404> (bumpers)
- <https://www.pololu.com/product/4760> (Encoder)
- <https://www.pololu.com/product/2386/specs> (Motors)
- https://www.amazon.com/Blomiky-1500mAh-Battery-m4-fpv250-Shredder/dp/B07DNYC78Q/ref=sr_1_16?crd=3LVPIPSVDW1XW&keywords=lipo+battery+1500mah&qid=1580866644&refinements=p_85%3A2470955011&rnid=2470954011&rps=1&s=toys-and-games&sprefix=lipo+battery+15%2Ctoys-and-games%2C139&sr=1-16 (Battery)
- <https://www.pololu.com/product/2450> (IR sensor)
- <https://www.adafruit.com/product/2927> (motor driver)
- <https://www.pololu.com/product/989/pictures> (motor mounts)

➤ <https://www.adafruit.com/product/4516> (microcontroller)

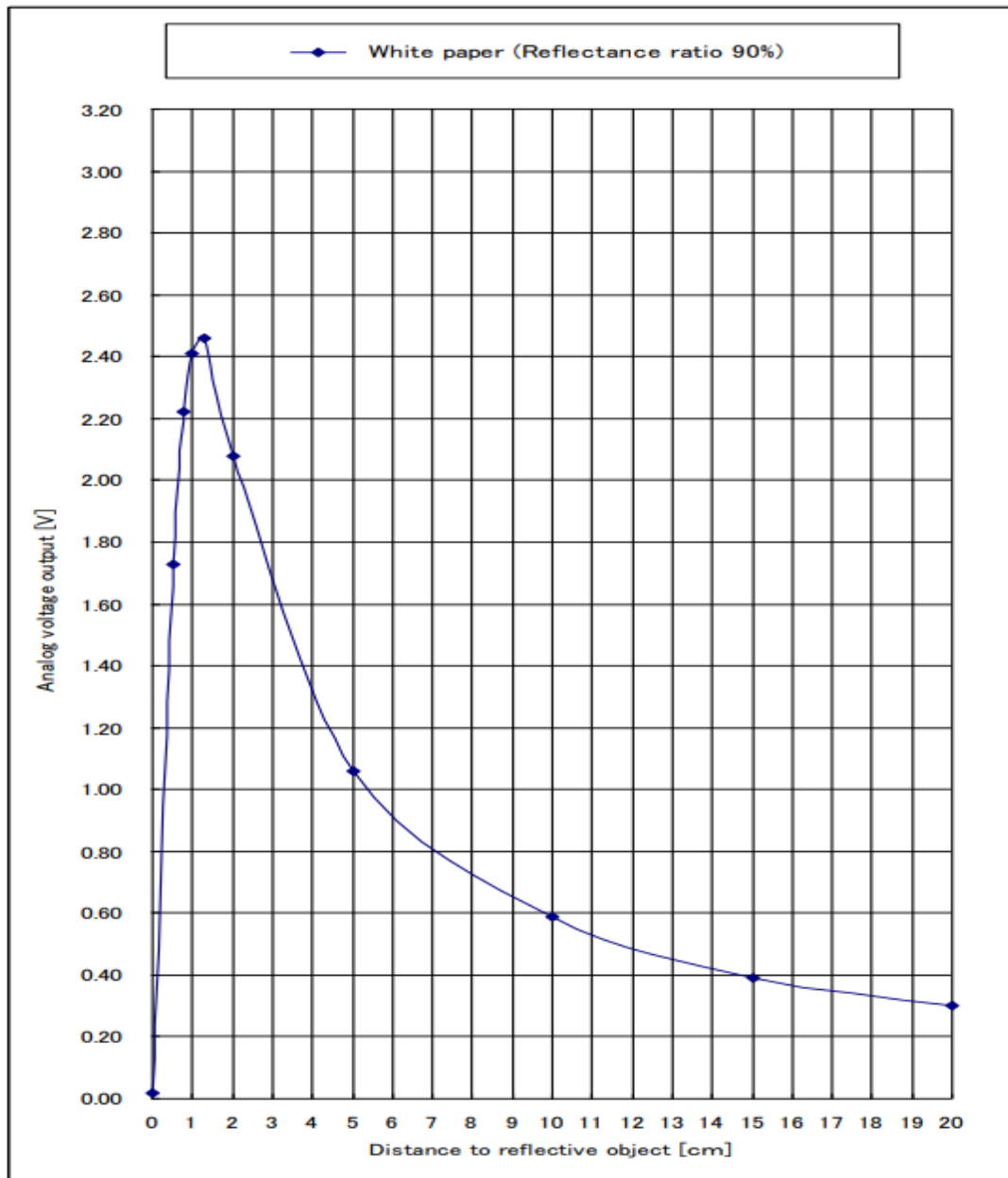
Pololu Items #997, #2386 (150:1 Micro Metal Gearmotor HP 6V) Performance at 6 V

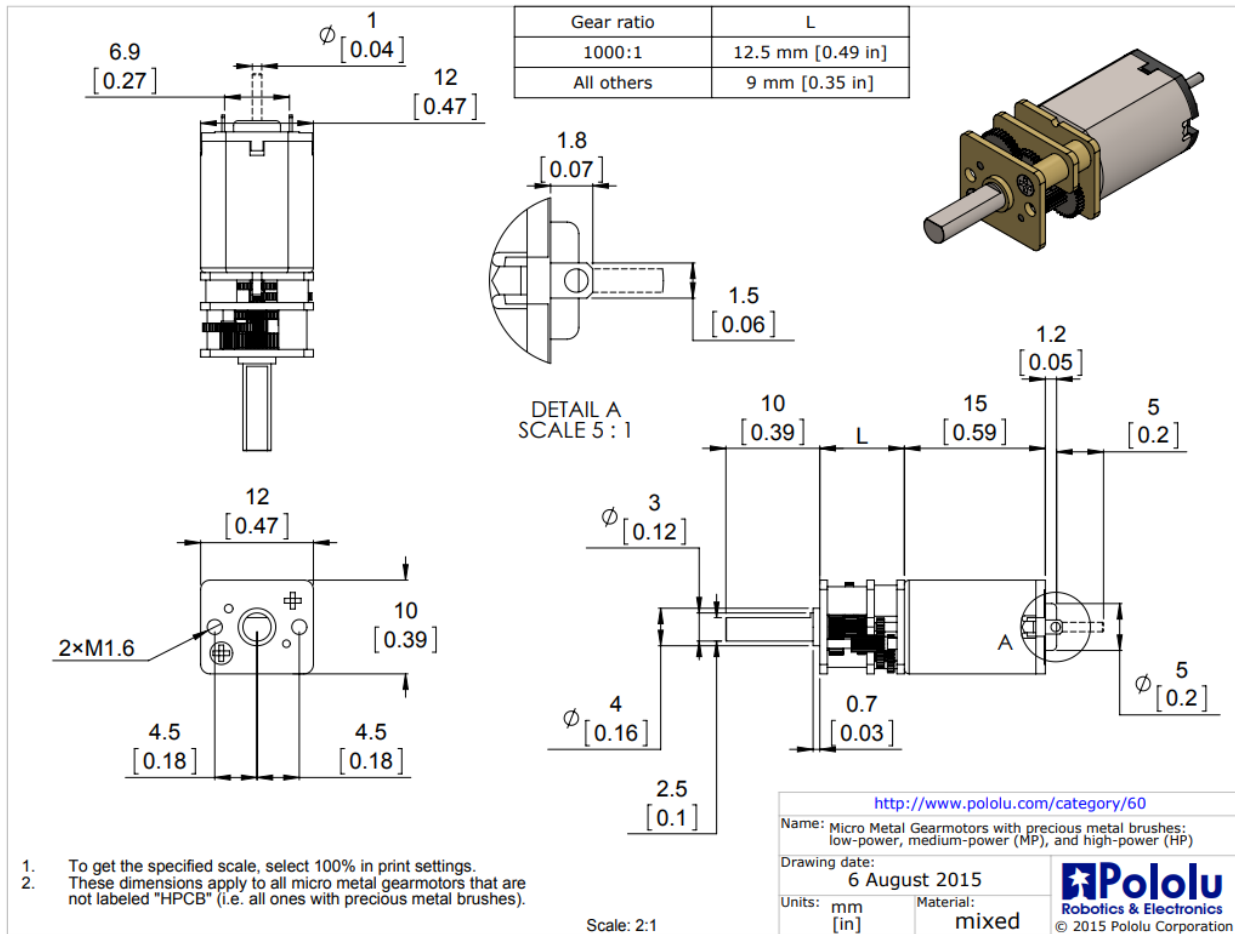


Motor datasheet

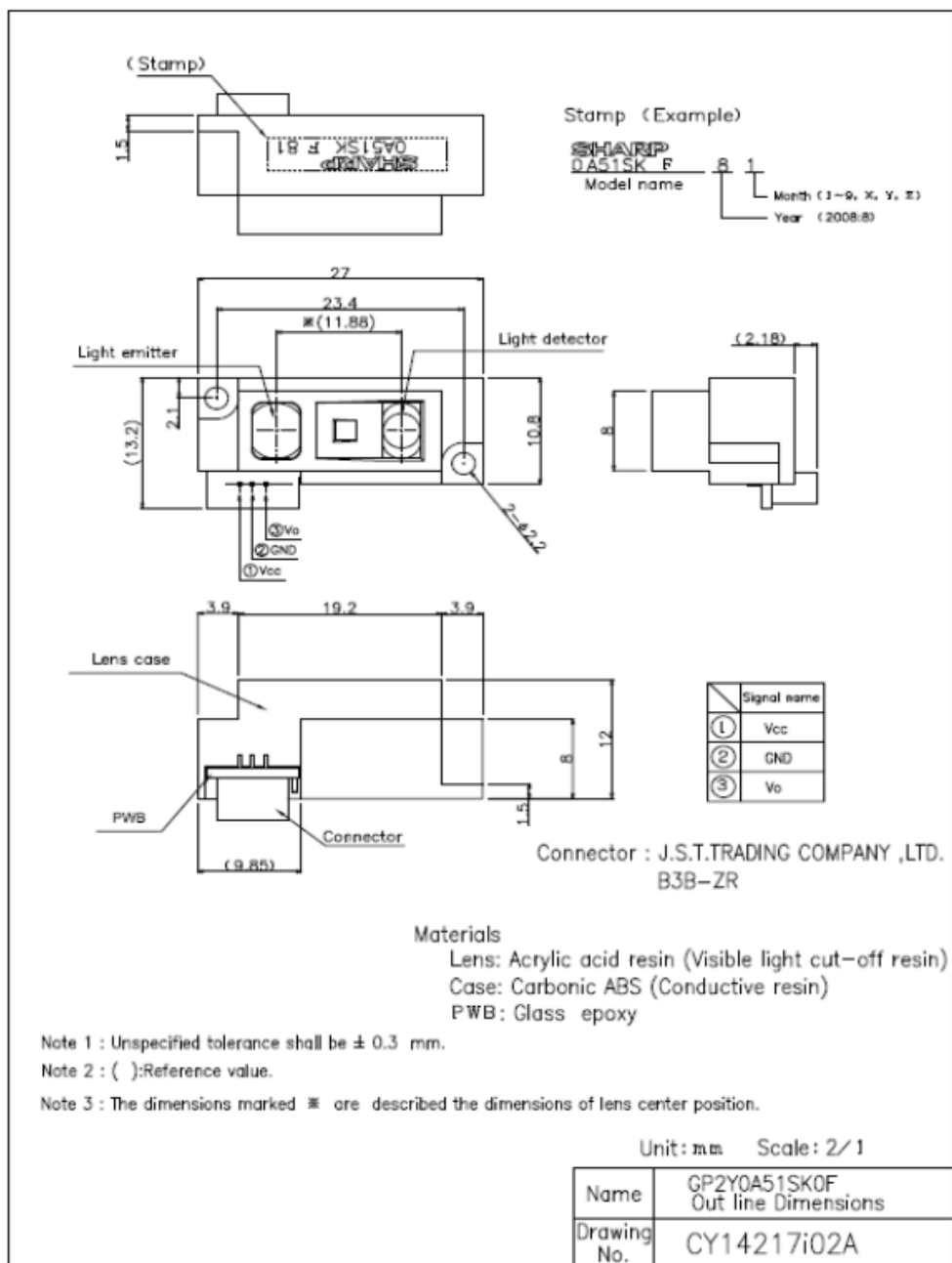
■Supplements

- Example of output distance characteristics

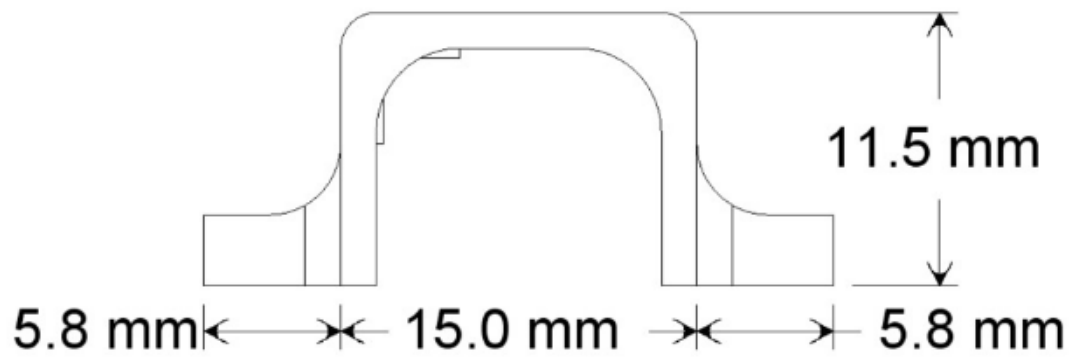
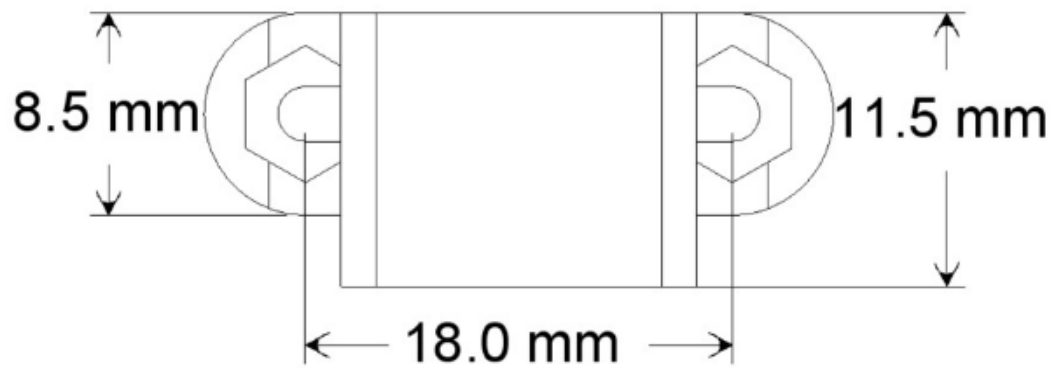




Motor dimensions

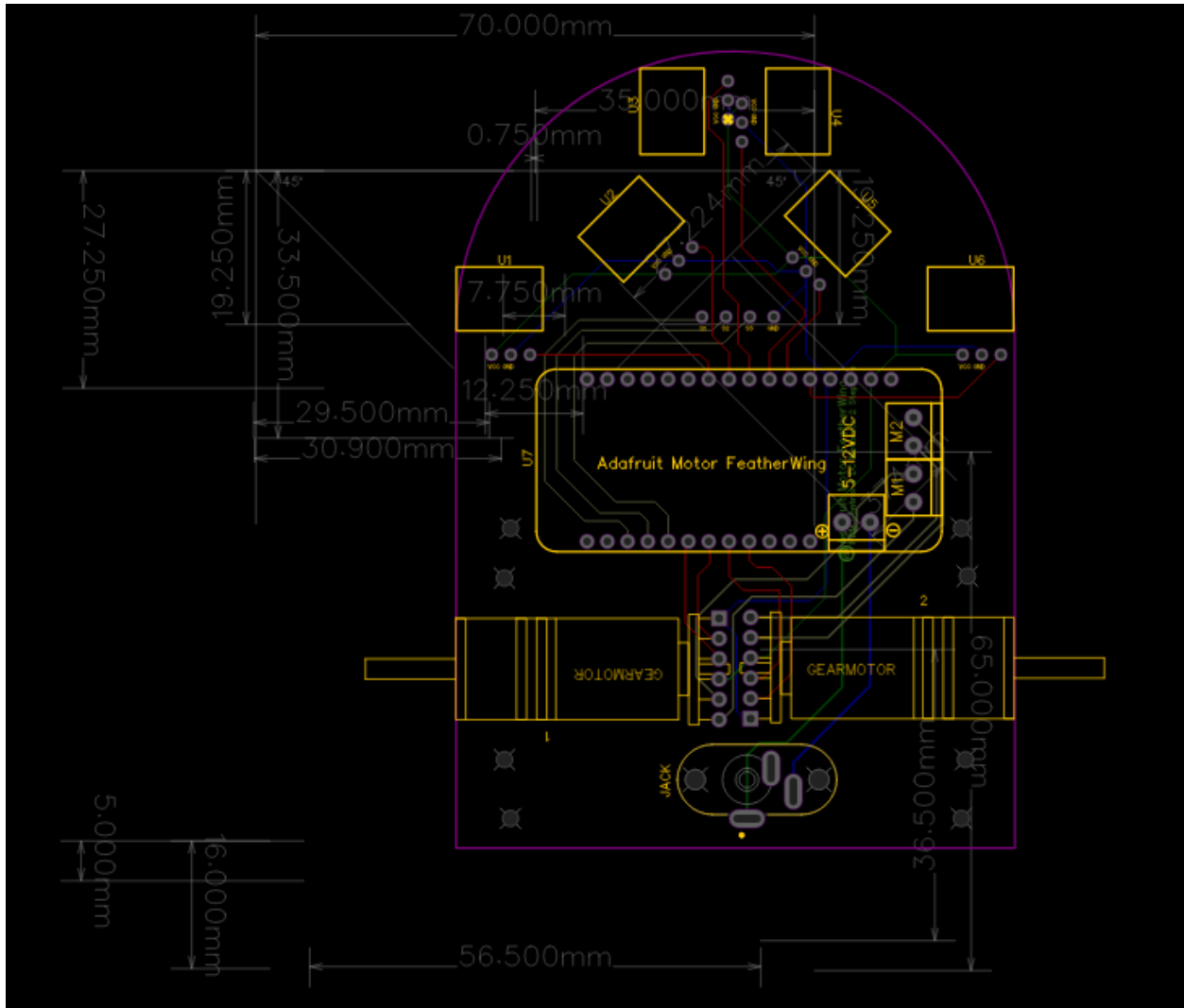


Sheet No.: OP13007EN

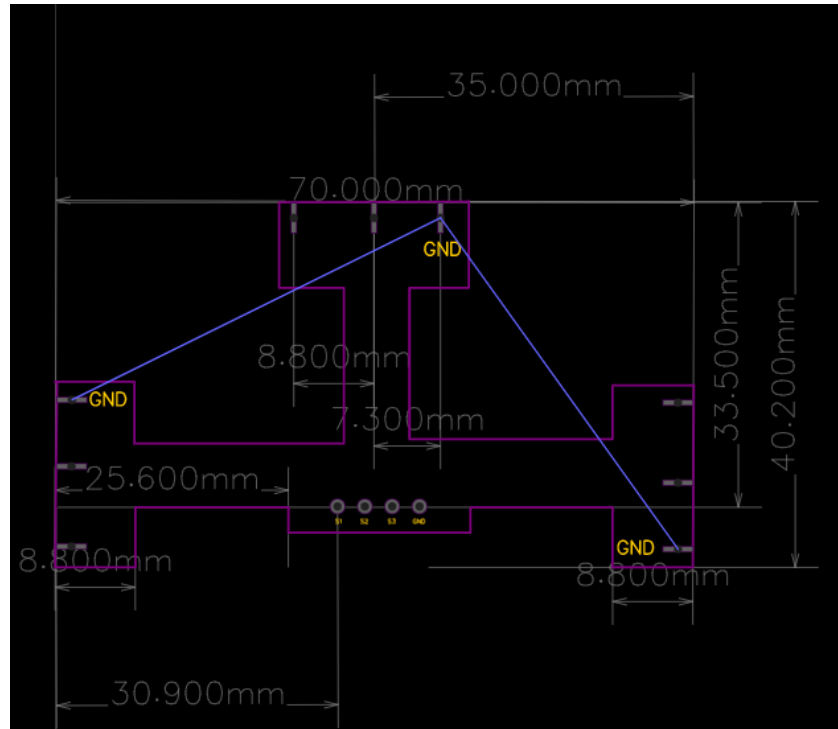


Dimensions of the metal gear motor mounting bracket.

Motor mount dimensions



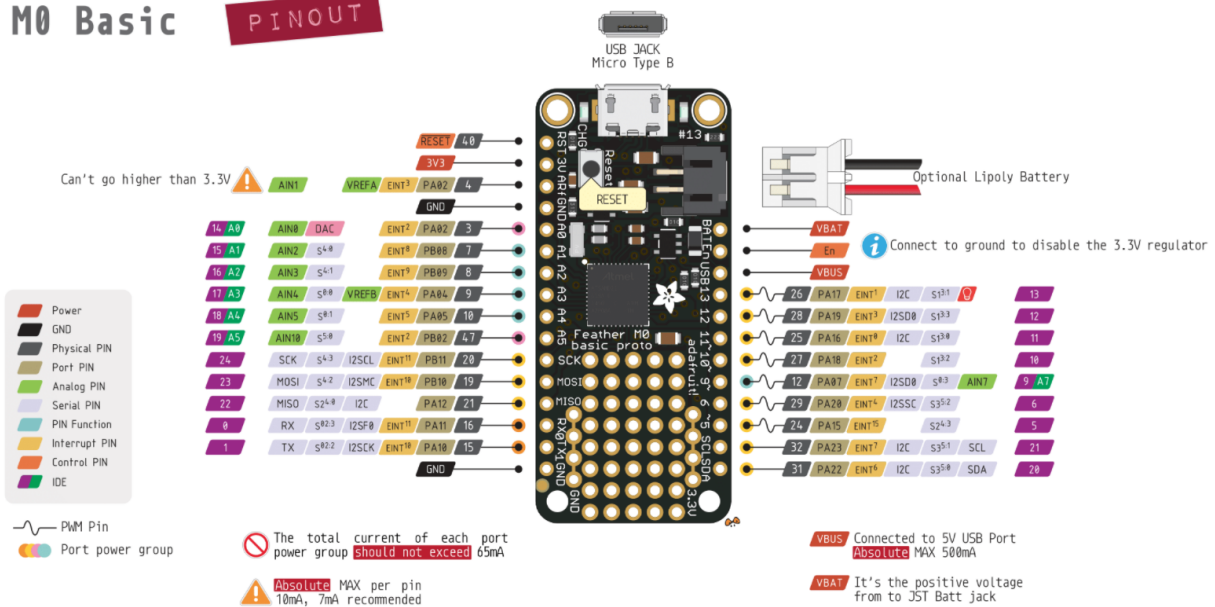
Primary PCB dimensions



Secondary PCB dimensions

M0 Basic

PINOUT



<https://learn.adafruit.com/assets/46244>

Microcontroller Pinouts

Table 6-1 Electrical Conductor Spacing

Voltage Between Conductors (DC or AC Peaks)	Minimum Spacing						
	Bare Board				Assembly		
	B1	B2	B3	B4	A5	A6	A7
0-15	0.05 mm [0.00197 in]	0.1 mm [0.0039 in]	0.1 mm [0.0039 in]	0.05 mm [0.00197 in]	0.13 mm [0.00512 in]	0.13 mm [0.00512 in]	0.13 mm [0.00512 in]
16-30	0.05 mm [0.00197 in]	0.1 mm [0.0039 in]	0.1 mm [0.0039 in]	0.05 mm [0.00197 in]	0.13 mm [0.00512 in]	0.25 mm [0.00984 in]	0.13 mm [0.00512 in]
31-50	0.1 mm [0.0039 in]	0.6 mm [0.024 in]	0.6 mm [0.024 in]	0.13 mm [0.00512 in]	0.13 mm [0.00512 in]	0.4 mm [0.016 in]	0.13 mm [0.00512 in]
51-100	0.1 mm [0.0039 in]	0.6 mm [0.024 in]	1.5 mm [0.0591 in]	0.13 mm [0.00512 in]	0.13 mm [0.00512 in]	0.5 mm [0.020 in]	0.13 mm [0.00512 in]
101-150	0.2 mm [0.0079 in]	0.6 mm [0.024 in]	3.2 mm [0.126 in]	0.4 mm [0.016 in]	0.4 mm [0.016 in]	0.8 mm [0.031 in]	0.4 mm [0.016 in]
151-170	0.2 mm [0.0079 in]	1.25 mm [0.0492 in]	3.2 mm [0.126 in]	0.4 mm [0.016 in]	0.4 mm [0.016 in]	0.8 mm [0.031 in]	0.4 mm [0.016 in]
171-250	0.2 mm [0.0079 in]	1.25 mm [0.0492 in]	6.4 mm [0.252 in]	0.4 mm [0.016 in]	0.4 mm [0.016 in]	0.8 mm [0.031 in]	0.4 mm [0.016 in]
251-300	0.2 mm [0.0079 in]	1.25 mm [0.0492 in]	12.5 mm [0.4921 in]	0.4 mm [0.016 in]	0.4 mm [0.016 in]	0.8 mm [0.031 in]	0.8 mm [0.031 in]
301-500	0.25 mm [0.00984 in]	2.5 mm [0.0984 in]	12.5 mm [0.4921 in]	0.8 mm [0.031 in]	0.8 mm [0.031 in]	1.5 mm [0.0591 in]	0.8 mm [0.031 in]
> 500 See para. 6.3 for calc.	0.0025 mm /volt	0.005 mm /volt	0.025 mm /volt	0.00305 mm /volt	0.00305 mm /volt	0.00305 mm /volt	0.00305 mm /volt

B1 - Internal Conductors

B2 - External Conductors, uncoated, sea level to 3050 m [10,007 feet]

B3 - External Conductors, uncoated, over 3050 m [10,007 feet]

B4 - External Conductors, with permanent polymer coating (any elevation)

A5 - External Conductors, with conformal coating over assembly (any elevation)

A6 - External Component lead/termination, uncoated, sea level to 3050 m [10,007 feet]

A7 - External Component lead termination, with conformal coating (any elevation)

Conductor Spacing table

Table 5-1 Fabrication Considerations

Fabrication Design Assumptions	Benefits(★), Drawbacks(↓), Impacts of Not Following Assumptions(⊗), Other Comments(⊗)
Hole/Land Ratio: Land size at least 0.6 mm [0.024 in] greater than the hole size ¹	★Provides sufficient land area to prevent breakout, i.e., hole intersecting edge of land (insufficient annular ring) ↓ Large lands may interfere with minimum spacing
Teardrop at Connection of Run with Land	★Provides additional area to prevent breakout. ★May improve reliability by preventing cracking at land/run boundary in vibration or thermal cycling. ↓ May interfere with minimum space requirements
Board Thickness: 0.8 mm to 2.4 mm [0.031 in to 0.0945 in] typical (over copper)	⊗Thinner boards tend to warp & require extra handling with through-hole technology components. Thicker boards have lower yield because of layer to layer registration. Some components may not have long enough leads for thicker boards.
Board Thickness to Plated Hole Diameter: Ratios ≤ 5:1 are preferred ¹	★Smaller ratios result in more uniform plating in hole, easier cleaning of holes and less drill wander. ★Larger holes are less susceptible to barrel cracking.
Symmetry across Board Thickness: top half should be a mirror image of bottom half to achieve a balanced construction	⊗Asymmetrical boards tend to warp. ⊗The location of ground/power planes, the orientation of signal runs and the direction of the fabric weave affect board symmetry. ↓ Heavy copper areas should be distributed throughout the area of the board as well to minimize warp.
Board Size	★Smaller boards warp less and have better layer to layer registration. ↓ Foil lamination or floating layer lay-ups should be considered for large panels with small features ⊗Panel utilization determines cost.
Conductor Spacing: ≤0.1 mm [≤0.0039 in]	⊗Etchant fluid does not circulate efficiently in narrower spaces resulting in incomplete metal removal.
Circuit Feature (Conductor Width): ≤0.1 mm [≤0.0039 in]	⊗Smaller features are more susceptible to breakage and damage during etching.

¹ These fabrication considerations, although valuable, may not be practical for some vias. Those vias which have small pad diameters cannot have 0.6 mm [0.024 in] of land size larger than the hole as this violates the board thickness to plated hole (aspect ratio) recommendation. When geometry considerations require small pads, the aspect ratio issue becomes paramount and the annular ring issue should be handled by exception.

Fabrication considerations (conductor spacing listed)

Test codes

Configuration file (Credited to - Luka Mgaloblishvili, Ali Khan)

```
import board
import analogio
from ulab import numpy as np
from adafruit_motorkit import MotorKit
import rotaryio
from src.encoder import Encoder
```

```
kit = MotorKit()
# Motors Configured:
l_motorKit = kit.motor1
r_motorKit = kit.motor2
#test_motorKit = kit.motor4
```

```
# Encoder Pins Configured:
r_encA = board.D13
r_encB = board.D12
l_encA = board.D10
l_encB = board.D11
```

```
# IR Sensor Pins Configured:
lAng_ir = analogio.AnalogIn(board.A4)
rAng_ir = analogio.AnalogIn(board.A1)
l_ir = analogio.AnalogIn(board.A3)
r_ir = analogio.AnalogIn(board.A2)
lFwd_ir = analogio.AnalogIn(board.A5)
rFwd_ir = analogio.AnalogIn(board.A0)
```

Motor Code (Credited to - Luka Mgaloblishvili, Ali Khan)

```
#Motor Code:
encA = rotaryio.IncrementalEncoder(config.l_encA, config.l_encB)
encB = rotaryio.IncrementalEncoder(config.r_encA, config.r_encB)
test_motor1 = Motor(config.l_motorKit, encA)
test_motor2 = Motor(config.r_motorKit, encB)
```

```
while True:
    test_motor1.run(0.6)
    test_motor2.run(0.6)
    time.sleep(1)
    print((test_motor1.get_speed(1), test_motor2.get_speed(1),))
```

```
#IGNORE
speed = 0
actual_speed1 = 0
actual_speed2 = 0
time_passed = 0
readings = 0
while readings < iterations:
    while speed < 1:
        test_motor1.run(speed)
        test_motor2.run(speed)
        time.sleep(interval)
        time_passed += interval
        actual_speed1 = test_motor1.get_speed(interval)
        actual_speed2 = test_motor2.get_speed(interval)
        print("INPUT SPEED:")
        print(speed)
        print("ACTUAL LEFT:")
        print((actual_speed1, actual_speed2,))
        # print((actual_speed1, actual_speed2, speed,))

        speed += 0.1

    speed = 1
```

```

while speed > 0:
    test_motor1.run(speed)
    test_motor2.run(speed)
    time.sleep(interval)
    time_passed += interval
    actual_speed = test_motor1.get_speed(interval)
    print("INPUT SPEED:")
    print(speed)
    print((actual_speed1, actual_speed2,))
    # print((actual_speed1, actual_speed2, speed,))
    speed -= 0.1

readings += 1

```

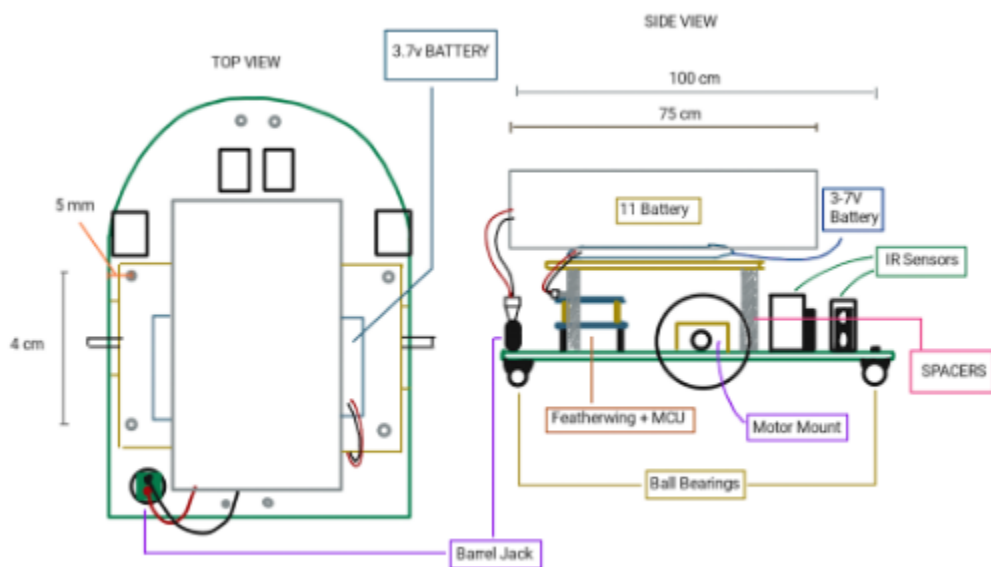
IR sensor Code (Credited to - Edwin Garcia)

```

#IR Sensor Code
lAng = IR_sensor(config.lAng_ir)
rAng = IR_sensor(config.rAng_ir)
l = IR_sensor(config.l_ir)
r = IR_sensor(config.r_ir)
lFwd = IR_sensor(config.lFwd_ir)
rFwd = IR_sensor(config.rFwd_ir)

while True:
    print((lFwd.read(), lAng.read(), l.read(),))
    time.sleep(1)

```



Rough Sketch of Assembled Micromouse